



INDIVIDUAL PROJECT

CHENNAI MATHEMATICAL INSTITUTE

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Project Title

Information & Coding Theory

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Chapter 1

Introduction

Hello!!

1.1 Objectives

Information theory, Coding theory and Probabilistic methods are very much important in Combinatorics, Computer science and modern technologies. So our **goal** is to learn the basics of these topics and how to apply these into real life. Our objective is to use these theories in our real life.

1.2 Challenges

I know probability theory but don't have ideas about information theory, coding theory, probabilistic methods. So I have to start from very much basics of these topics. Also here, some advanced facts of algebra and Jensen's inequality are used in this project. So some readers can face problems with that.

1.3 Contributions

I have used various internet sources to do this project. I have converted the project into pdf using Overleaf. Many digital copies of books are also used in this project. My Supervisor gave much study materials which are great and these materials helped me to complete the project.

Chapter 2

Background

The Probabilistic Method has recently been developed intensively and became one of the most powerful and widely used tools applied in Combinatorics. One of the major contribution of this is in the role of **randomness in Theoretical Computer Science, Circuit Complexity , Computational Geometry and De-randomization of randomized algorithms**.

The interplay between Discrete Mathematics and Computer Science suggests an **algorithmic** point of view in the study of the Probabilistic Method in Combinatorics. **Paul Erdos** initiated the basic probabilistic method and made many conjectures that stimulated a big portion of the research in the area. Interest in the concept of information grew directly from the creation of the telegraph and telephone. In 1844 the American inventor **Samuel F.B. Morse** built a telegraph line and encountered many electrical problems when he sent signals through buried transmission lines, but inexplicably he encountered fewer problems when the lines were suspended on poles. In 1924 **Harry Nyquist** realized that communication channels had maximum data transmission rates, and he derived a formula for calculating these rates in finite bandwidth noiseless channels. A key step in **Claude Shannon**'s work was a theory, communication signals must be treated in isolation from the meaning of the messages that they transmit. This view is in sharp contrast with the common conception of information, in which meaning has an essential role. Shannon also realized that the amount of knowledge conveyed by a signal is not directly related to the size of the message. Shannon's work defined communication channels and showed how to assign a capacity to them, not only in the theoretical sense where no interference, or noise, was present but also in practical cases where real channels were subjected to real noise. Shannon produced a formula that showed how the **bandwidth** of a channel (that is, its theoretical signal capacity) and its **signal-to-noise ratio** (a measure of interference) affected its capacity to carry signals. In doing so he was able to suggest strategies for maximizing the **capacity** of a given channel and showed the limits of what was possible with a given technology. After the successful transmission still there was a question how to transmit the information correctly so that the receiver gets the exact information what we want to transmit. This question opens a new phase of information theory called coding theory.

Chapter 3

PROJECT

3.1 Information Theory

3.1.1 Introduction

In modern technology, the transmission, processing, and storage of information play the main role. For example, information transmission between the spacecraft and the control center. For the transmission of information, we replace each part of information with some specific symbol and create a sign sequence for an information. This process is called coding and the symbols are called codewords. For example, **english alphabets** can be coded using **binary digits**. There are 26 letters. Using **5 digit binary numbers** we can code all letters. The method of encoding depends on the type of transmission. For example, for the transmission of telegrams, the message is coded in the **Morse code**, in which sequences of dots and dashes are assigned to letters. The IV transmitter codes the picture by decomposing it into **small dots**.

Fact 1: If there are n distinct characters then we need $\log_2 n$ bits of information.

3.1.2 Loomis-Whitney inequality

There are n points in 3-dim space. Suppose these points have n_1, n_2, n_3 distinct projections on XY, XZ, YZ plane respectively. Then $n^2 \leq n_1 \cdot n_2 \cdot n_3$.

Proof 1: There are n points, so we need $\log_2 n$ bits of information to describe one of those n points. Similarly we need $\log_2 n_1, \log_2 n_2, \log_2 n_3$ bits of information to describe the projections in XY, XZ, YZ planes respectively. From projections we can get each coordinate twice. So $2 \log_2 n \leq \log_2 n_1 + \log_2 n_2 + \log_2 n_3 \Rightarrow n^2 \leq n_1 \cdot n_2 \cdot n_3$.

3.1.3 Optimal encoding

In optimal coding we minimize the number of bits transmitted on average. For example: Let we have 6 letters a,b,c,d,e,f. X is r.v. $P(a) = 0.1, P(b) = 0.2, P(c) = 0.05, P(d) = 0.25, P(e) = 0.2, P(f) = 0.2$. We denote a by 1, b by 10, c by 11, d by 100, e by 101, f by 110. Then $E(\text{no. of bits}) = 1 \cdot 0.1 + 2 \cdot 0.2 + 2 \cdot 0.05 + 3 \cdot 0.25 + 3 \cdot 0.2 + 3 \cdot 0.2 = 2.55$ bits on average. This optimal value is denoted by $T(x)$.

For encoding in binary digits we have a similar structure, i.e., **binary tree**. Edge from one node to its left child is denoted by 0 and edge from that node to its right child is denoted by 1. So from root to one node we have **sequence of binary digits**. We replace that node by that sequence in encoding.

3.1.4 Huffman Coding

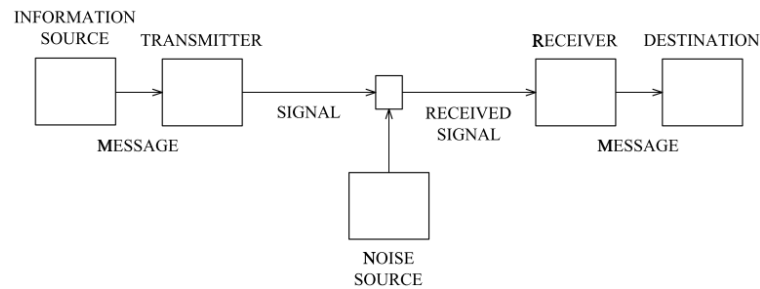
Find the smallest codeword. Then denote the codewords s.t. some codeword is the prefix of that codeword. For example, a,b,c 3 letters. $P(a) = 0.5, P(b) = 0.25, P(c) = 0.25$. Replace a by 0, b by 10 and c by 11 then we have a binary tree structure where root has left child a, the right child has left child b and right child c. Then $T(x) = 0.5 \cdot 1 + 0.25 \cdot 2 + 0.25 \cdot 2 = 1.5$. Here length of a $= l_a = \lceil \log_2 \frac{1}{P(a)} \rceil = 1, l_b = \lceil \log_2 \frac{1}{P(b)} \rceil = 2, l_c = \lceil \log_2 \frac{1}{P(c)} \rceil = 2$

In general $T[X] = l_a P(a) + l_b P(b) + l_c P(c)$

In general let X is r.v. which takes values $1, 2, \dots, k$ with prob $p_1, p_2, \dots, p_k$. Then entropy of X $= H[X] = \sum_{i=1}^n p_i \log_2(\frac{1}{p_i})$

3.1.5 The Science of Information

In a communication system, we try to convey information from one point to another, very often in a noisy environment. Consider some conditions: let a person wants to send facsimiles regularly and wants as much information as possible in one page. He can choose the font size such that as much information he wants, can fit in one page. But there is another condition: fax machine has finite resolution. If the font size is too small, then the characters may not be recognizable on the facsimile and although some characters on the facsimile may not be recognizable, the recipient can still figure out the words from the context provided that the number of such characters is not excessive. In other words, it is not necessary to choose a font size such that all the characters on the facsimile are recognizable almost surely. Then what is the maximum amount of meaningful information which can be conveyed on one page of facsimile? Information is an abstract concept, hard to quantify in general. This is especially the case if human factors are involved when the information is utilized. For example: when we play a CD, we receive musical information from loudspeakers. We enjoy this information because it arouses certain kinds of emotion within ourselves. While we receive the same information every time we play the same piece of music, the kinds of emotions aroused may be different from time to time because they depend on our mood at that particular moment. In other words, we can derive utility from the same information every time in a different way.



Bell Telephone Laboratories scientist **Claude E. Shannon** published a paper entitled “The Mathematical Theory of Communication”. In his paper, the model of a point-to-point communication system depicted. In this model, a message is generated by the information source. The message is converted by the transmitter into a signal which is suitable for transmission. In the course of transmission, the signal may be contaminated by a noise source, so that the received signal may be different from the transmitted signal. Based on the received signal, the receiver then makes an estimate of the message and deliver it to the destination. In this model, one is only concerned about whether the message generated by the source can be delivered correctly to the receiver without worrying about how the message is actually used by the receiver. Shannon introduced two fundamental concepts about ‘information’ from the communication point of view. First, information is **uncertainty**. If some information is deterministic then there is no uncertainty. Second, information to be transmitted is **digital**. This means that the information source should first be converted into a stream of 0’s and 1’s called bits and the remaining task is to deliver these bits to the receiver correctly with no reference to their actual meaning. This is the foundation of all modern digital communication systems. Shannon also proved two important theorems. 1st theorem, **Source Coding Theorem** introduces **entropy**, fundamental measure of information which characterizes the minimum rate of a source code representing an information source essentially free of error. The source coding theorem is the theoretical basis for **lossless data compression**. 2nd theorem, **Channel Coding Theorem** concerns communication through a noisy channel. Each noisy channel has a parameter called **capacity** which is strictly positive except for very special channels, such that information can be communicated reliably through the channel as long as the information rate is less than the capacity. These two theorems, which give fundamental limits in point-to-point communication, are the two most important results in information theory.

3.1.6 Entropy

The entropy of a random variable is a measure of the **uncertainty** of the random variable; it is a measure of the amount of information required on the average to describe the random variable.

Postulate 1: A larger number of potential outcomes have larger uncertainty.
Larger the number of outcomes, harder to it is to predict and hence the greater the uncertainty.

Postulate 2: The relative likelihood of each outcome determines the uncertainty.
The uncertainty $H[X]$ as a function of the probability distribution $p = \{p_1, p_2, \dots, p_A\}$ dictating the frequency of all A outcomes. Thus, by the first postulate $\frac{dH}{dA} > 0$ since the uncertainty grows monotonically as the number of outcomes increases (We assume H is a continuous function).

Postulate 3: The weighted uncertainty of independent events must sum.

This final postulate was a stroke of genius recognized by Claude Shannon. In the case of rolling two dice, this means that the total uncertainty of rolling two independent dice must equal the sum of the uncertainties for each die alone. In other words, if the uncertainty of each die is H_1 and H_2 respectively, then the total uncertainty of rolling both die simultaneously must be $H_1 + H_2$. Weighted refers to the fact that the uncertainties should be weighted by the probability of occurrence. For the probability distribution p the only function that matches these intuitions is:

$$H[p] = -\sum_{i=1}^A p_i \log_2(p_i)$$

Fact 2: If the prob are measured based on n outcomes then $H[X] \leq \log_2 n$.

Derivation of Entropy function: Consider two separate, independent, uniform probability distributions with x and y elements respectively. The composition law requires that:

$H(x) + H(y) = H(xy)$, where $H(x)$ refers to the entropy of a uniform distribution with x outcomes. This is equivalent of, the uncertainty of simultaneously rolling a x -sided and a y -sided die is equal to the sum of the uncertainties for each die alone. In the single die case there exist x and y equally probable outcomes respectively, and in the simultaneous case, there exist xy equally probable outcomes. To derive the uniform form for H we take the derivative with respect to each variable: $\frac{dH(x)}{dx} = y \frac{dH(xy)}{dx}$ and $\frac{dH(y)}{dy} = x \frac{dH(xy)}{dy} \Rightarrow x \frac{dH(x)}{dx} = y \frac{dH(y)}{dy}$.

Each side of this equation is solely a function of an arbitrary choice of variables, thus equality can only hold for all x and y if and only if both sides equal a constant k , $x \frac{dH(x)}{dx} = k \Rightarrow H(x) = k \log x + c$, where c is a constant. We set $c = 0$ because there is zero uncertainty when only a single outcome is possible. The first postulate requires that $\frac{dH(x)}{dx} > 0 \Rightarrow k > 0$. We select base 2 logarithms to provide the familiar units of bits, resulting in the final form of the equation, $H(x) = \log_2 x$.

For non-uniform case, take $N = \sum_i n_i$ and denote: $p_i = \frac{n_i}{N}$ then: $\sum_{i=1}^A p_i H(n_i) = \sum_{i=1}^A p_i (\log_2 n_i)$
 $\Rightarrow \log N = H(p) + \sum_{i=1}^A p_i (\log_2 n_i) \Rightarrow H(p) = \log N - \sum_{i=1}^A p_i (\log_2 n_i) = -\sum_{i=1}^A p_i (\log_2 p_i)$

3.1.7 Shannon's Theorem

$$H[X] \leq T[X] \leq H[X] + 1$$

proof is obvious from the formulas of $T[X], H[X]$

Shannon's Theorem for Entropy: $H(X), H(Y) \leq H(X, Y) \leq H(X) + H(Y)$

3.1.8 Another proof of Loomis-Whitney inequality

Using entropy. $P \equiv (X, Y, Z)$ be a coordinates of P . $P_1 \equiv (X, Y), P_2 \equiv (X, Z), P_3 \equiv (Y, Z)$.
 $H[P] = H[XYZ] = H[X] + H[Y|X] + H[Z|XY], H[XY] = H[X] + H[Y|X], H[YZ] = H[Y] + H[Z|Y], H[ZX] = H[X] + H[Z|X]$ Then $2H[P] \leq H[P_1] + H[P_2] + H[P_3] \Rightarrow n^2 \leq n_1 n_2 n_3$

3.1.9 Joint Entropy and Conditional Entropy

Conditional Entropy: Conditioning reduces entropy. **Formula:** $H[Y|X] = H[XY] - H[X]$
 $H[Y|X] = 0 \iff Y$ is determined by X , i.e., there is a deterministic function f s.t. $f(X) = Y$.
 $H[X_1, X_2, \dots, X_n] \leq H[X_1] + H[X_2] + \dots + H[X_n]$ equality holds when $X_1, X_2, \dots, X_n$ are mutually independent r.v. as $H[X_1, X_2] = H[X_1] + H[X_2]$ when X_1, X_2 are independent r.v.

Fact 3: Entropy is maximum when all outcomes are equally likely.

Joint Entropy: The joint entropy $H(X, Y)$ of a pair of discrete random variables (X, Y) with a joint distribution $p(x, y)$ is defined as: $H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(x, y))$
Conditional entropy $H(Y|X) = \sum_{x \in X} p(x) H(Y|X = x) = -\sum_{x \in X} p(x) \sum_{y \in Y} p(y|x) \log(p(y|x)) =$

$$-\sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(y|x))$$

Formula: $H[Y|X] \leq H[Y]$ and $H[Z|XY] \leq H[Z|X]$

Chain Rule: $H(X, Y) = H(X) + H(Y|X)$

$$\begin{aligned} \text{Pf: } H(X, Y) &= -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(x, y)) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(x)p(y|x)) \\ &= -\sum_{x \in X} p(x) \log(p(x)) - \sum_{x \in X} \sum_{y \in Y} p(x, y) \log(p(y|x)) = H(X) + H(Y|X) \end{aligned}$$

Corollary: $H(X, Y|Z) = H(X|Z) + H(Y|X, Z)$

In general, For a sequence of random variables $X_1, X_2, \dots, X_n$ denote $X_{<i} = (X_1, X_2, \dots, X_{i-1})$ then $H(X_1, X_2, \dots, X_n) = \sum_i H(X_i|X_{<i}) = H(X_1) + H(X_2|X_1) + H(X_3|X_1, X_2) + \dots + H(X_n|X_1, X_2, \dots, X_{n-1})$

Property: For random variables X and Y and a function $f(Y)$, $H(Y|X) \leq H(X|f(Y))$

Pf: The sequence $X \rightarrow Y \rightarrow f(Y)$ forms a Markov chain. Thus by the data processing inequality: $I(X : f(Y)) \leq I(X : Y) \Rightarrow H(X) - H(X|f(Y)) \leq H(X) - H(X|Y) \Rightarrow H(X|Y) \leq H(X|f(Y))$

3.1.10 Relative Entropy and Mutual Information

The relative entropy is a measure of the distance between two distributions. In statistics, it arises as an expected logarithm of the likelihood ratio. The relative entropy $D(p||q)$ is a measure of the **inefficiency** of assuming that the distribution is q when the true distribution is p . For example, if we knew the true distribution p of the random variable, we could construct a code with average description length $H(p)$. If, instead, we used the code for a distribution q , we would need $H(p) + D(p||q)$ bits on the average to describe the random variable.

The relative entropy or Kullback–Leibler distance between two probability mass functions $p(x), q(x)$ is defined as: $D(p||q) = \sum_{x \in X} p(x) \log(\frac{p(x)}{q(x)}) = E_p \log(\frac{p(X)}{q(X)})$

Conventionally, $0 \log(\frac{0}{0}) = 0, 0 \log(\frac{0}{q}) = 0, q \log(\frac{p}{0}) = \infty$

Def: Consider two random variables X and Y with a joint probability mass function $p(x, y)$ and marginal probability mass functions $p(x), p(y)$. The mutual information $I(X; Y)$ is the relative entropy between the joint distribution and the product distribution $p(x)p(y)$:

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log(\frac{p(x, y)}{p(x)p(y)}) = D(p(x, y)||p(x)p(y)) = E_{p(x, y)} \log(\frac{p(X, Y)}{p(X)p(Y)})$$

Example: Let $X = \{0, 1\}$ and consider two distributions p and q on X . Let $p(0) = 1 - r, q(0) = 1 - s, p(1) = r, q(1) = s$. Then

$$D(p||q) = (1 - r) \log(\frac{1-r}{1-s}) + r \log(\frac{r}{s}) \text{ and } D(q||p) = (1 - s) \log(\frac{1-s}{1-r}) + s \log(\frac{s}{r})$$

If $r = s$ then $D(p||q) = D(q||p) = 0$. If $r = \frac{1}{2}, s = \frac{1}{4}$ then $D(p||q) = 0.2075$ bit, $D(q||p) = 0.1887$ bit. So $D(p||q) \neq D(q||p)$

Relationship between Entropy and Mutual Information:

$$\begin{aligned} I(X; Y) &= \sum_{x \in X} \sum_{y \in Y} p(x, y) \log(\frac{p(x, y)}{p(x)p(y)}) = \sum_{x, y} p(x, y) \log(\frac{p(x, y)}{p(x)}) - \sum_{x, y} p(x, y) \log(p(y|x)) \\ &= \sum_{x, y} p(x, y) \log(p(x|y)) = H(X) - H(X|Y). \text{ By symmetry, } I(X; Y) = H(Y) - H(Y|X) \Rightarrow \\ I(X; Y) &= H(X) + H(Y) - H(X, Y). \text{ So } I(X; X) = H(X) - H(X|X) = H(X) \end{aligned}$$

Theorem: $I(X; Y) = I(Y; X) = H(X) - H(X|Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X, Y)$, $I(X; X) = H(X)$

Conditional Mutual Information: The conditional mutual information of random variables X and Y given Z is defined by $I(X; Y|Z) = H(X|Z) - H(X|Y, Z)$

Chain Rule for Information: $I(X_1, X_2, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y|X_{i-1}, \dots, X_2, X_1)$

$$\begin{aligned} \text{Pf: } I(X_1, X_2, \dots, X_n; Y) &= H(X_1, X_2, \dots, X_n) - H(X_1, X_2, \dots, X_n|Y) = \sum_{i=1}^n H(X_i|X_{i-1}, \dots, X_2, X_1) - \\ \sum_{i=1}^n H(X_i|X_{i-1}, \dots, X_2, X_1, Y) &= \sum_{i=1}^n I(X_i; Y|X_{i-1}, \dots, X_2, X_1) \end{aligned}$$

Conditional Relative Entropy: For joint probability mass functions $p(x, y), q(x, y)$ the conditional relative entropy $D(p(y|x)||q(y|x))$ is the average of the relative entropies between the conditional probability mass functions $p(y|x), q(y|x)$ averaged over the probability mass function $p(x)$. $D(p(y|x)||q(y|x)) = \sum_x p(x) \sum_y p(y|x) \log(\frac{p(y|x)}{q(y|x)}) = E_{p(x, y)} \log(\frac{p(Y|X)}{q(Y|X)})$

Chain rule for relative entropy: $D(p(x, y)||q(x, y)) = D(p(x)||q(x)) + D(p(y|x)||q(y|x))$

Pf: $D(p(x, y)||q(x, y)) = \sum_{x,y} p(x, y) \log(\frac{p(x,y)}{q(x,y)}) = \sum_{x,y} p(x, y) \log(\frac{p(x)p(y|x)}{q(x)q(y|x)}) = \sum_{x,y} p(x, y) \log(\frac{p(x)}{q(x)}) + \sum_{x,y} p(x, y) \log(\frac{p(y|x)}{q(y|x)}) = D(p(x)||q(x)) + D(p(y|x)||q(y|x))$

3.1.11 Entropy & Sets

Random variables X, Y, Z	Finite sets A, B, C
Entropy $H(X)$	Cardinality $ A $
Joint variable (X, Y)	Union $A \cup B$
Mutual information $I(X : Y)$	Intersection cardinality $ A \cap B $
Conditional entropy $H(X Y)$	Set difference cardinality $ A \setminus B $
Conditional mutual information $I(X : Y Z)$	$ (A \cap B) \setminus C $
X, Y independent	A, B disjoint
X determined by Y	A a subset of B
X, Y conditionally independent relative to Z	$A \cap B \subset C$

Every (linear) inequality or identity about entropy (and related quantities, such as mutual information) specialises to a combinatorial inequality or identity about finite sets that is easily verified. Shannon inequality: $H(X, Y) \leq H(X) + H(Y) \Rightarrow |A \cup B| \leq |A| + |B|$ (union bound). Definition of mutual information becomes **Principle of Inclusion & Exclusion:** $|A \cup B| = |A| + |B| - |A \cap B|$. Data processing inequality: if X, Z are conditionally independent relative to Y then $I(X : Z) \leq I(X : Y)$, i.e., if A, C are disjoint outside of B then $|A \cap C| \leq |A \cap B|$. In general, $|A \cap C| \leq |A \cap B| + |(A \cap C) - B| \Rightarrow I(X : Y) + I(X : Z|Y) = I(X : Z) + I(X : Y|Z)$. A basic property of sets is that disjointness from a given set C is preserved by unions: $|(A \cup B) \cap C| \leq |A \cap C| + |B \cap C| \Rightarrow I(X, Y : Z) \leq I(X : Y) + I(Y : Z)$. But these reverse direction implications are wrong, i.e., entropy equation $\Rightarrow$ set equation, but set equation doesn't imply entropy equation. The basic issue here is that in the dictionary between entropy and combinatorics, there is no satisfactory entropy analogue of the notion of a triple intersection $A \cap B \cap C$. The issue only arises with three or more variables. **Counter-Example:** $X, Y \in F_2$ to be independent, uniformly distributed random elements of the finite field, $Z = X + Y$. One can check that each of X, Y is separately independent of Z , but the joint distribution (X, Y) determines Z , is not independent of Z .

3.1.12 Entropy & Vector Spaces

Random variables X, Y, Z	Subspaces U, V, W
Entropy $H(X)$	Dimension $\dim(U)$
Joint variable (X, Y)	Sum $U + V$
Mutual information $I(X : Y)$	Dimension of intersection $\dim(U \cap V)$
Conditional entropy $H(X Y)$	Dimension of projection $\dim(\pi_V(U))$
Conditional mutual information $I(X : Y Z)$	$\dim(\pi_W(U) \cap \pi_W(V))$
X, Y independent	U, V transverse ($U \cap V = \{0\}$)
X determined by Y	U a subspace of V
X, Y conditionally independent relative to Z	$\pi_W(U), \pi_W(V)$ transverse.

Let $f : S \rightarrow F$ be a random linear functional on S , selected uniformly. Every subspace U of S then gives rise to a random variable $X = X_U : U \rightarrow F$ formed by restricting f to U . Its entropy can be easily computed to be $\dim(U) \log |F|$. The joint r.v. (X, Y) determines the random linear function f on the union $U \cup V$ on the two spaces, by linearity on the Minkowski sum: $H(X, Y) = \dim(U + V) \log |F|$ and $H(X|Y) = \dim(\pi_V(U)) \log |F|$, where $\pi_V : S \rightarrow S/V$ is the quotient map. Every linear inequality or equality that is valid for the information-theoretic quantities discussed above, is automatically valid for the linear algebra counterparts for subspaces of a vector space over a finite field. In fact, the requirement that the field be finite can be removed by applying the compactness theorem from logic (Los's Theorem on Ultraproducts).

Dimension inequality (Ingleton's inequality):

$$\dim(U \cap V) \leq \dim(\pi_W(U) \cap \pi_W(V)) + \dim(\pi_X(U) \cap \pi_X(V)) + \dim(W \cap X)$$

When 3 terms in RHS vanish, $\pi_W(U), \pi_W(V)$ are transverse, which implies $U \cap V \subseteq W$ and $U \cap V \subseteq X$. But X and W are transverse $\Rightarrow U$ and V are transverse. One can define: $Y = U \cap V$ and use $\dim(Y) \leq \dim(\pi_W(U) \cap \pi_W(V))$ to reduce to establish the inequality: $\dim(Y) \leq \dim(\pi_W(Y)) + \dim(\pi_X(Y)) + \dim(W \cap X) \dots$ (3). Now, $\dim(\pi_W(Y)) = \dim(Y) - \dim(W) + \dim(\pi_Y(W))$ and

$\dim(W \cap X) = \dim(W) + \dim(X) - \dim(W + X)$ as $\dim(W + X) \leq \dim(\pi_Y(W)) + \dim(\pi_X(Y)) + \dim(Y)$, this is true because $\dim(W + X) \leq \dim(\pi_Y(W) + \pi_X(X)) + \dim(Y)$.

The analogue of (3) is: $H(V) \leq H(V|Z) + H(V|W) + I(Z : W)$ is true. But the analogue of Ingleton's inequality is $I(X : Y) \leq I(X : Y|Z) + I(X : Y|W) + I(Z : W)$ is false in general. When all terms in RHS vanish, X, Y are conditionally independent relative to Z , relative to W and Z and W are independent, claim would be asserting X and Y are independent.

Counter-Example: Take Z, W independent uniform variables from F_2 and take $X = ZW$, $Y = (1 - Z)(1 - W)$. Once one condition on either Z or W , one of X, Y has positive conditional entropy and the other has zero entropy. So X, Y are independent relative to Z or W . and Z or W are independent of each other. But X and Y are not independent of each other.

Zhang-Yeung inequality:

$$I(X : Y) \leq 2I(X : Y|Z) + I(X : Z|Y) + I(Y : Z|X) + I(X : Y|W) + I(Z : W)$$

Logarithmically averaged Chowla conjecture: $|\sum_{n \leq x} \frac{\lambda(n)\lambda(n+1)}{n}| \leq \epsilon \log x$, whenever x is sufficiently large depending on $\epsilon > 0$, where λ is the Liouville function. If this inequality fails then there was a lower bound $I(X_H : Y_H) >> \epsilon^7 \frac{H}{\log H}$ on the mutual information between X_H and Y_H . R.v. X_H is the sign pattern of the form $(\lambda(n+1), \dots, \lambda(n+H))$. From translation invariance, this also gives the more general lower bound $I(X_{H,H_0} : Y_H) >> \epsilon^7 \frac{H}{\log H}$ for any H_0 , where $X_{H_0,H}$ denotes the shifted sign pattern $(\lambda(n+H_0+1), \dots, \lambda(n+H_0+H))$.

3.1.13 Jensen's Inequality and its Consequences

Jensen's Inequality: If f is convex function over interval (a, b) , then for every $x_1, x_2 \in (a, b)$ and $0 \leq \lambda \leq 1$: $f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$. For concave function $\leq$ becomes $\geq$. Equality holds only if $\lambda = 0$ or 1 .

Example: For any $p, p' \in [0, 1]$, $H(p + p' - pp') \leq (1 - p)H(p')$

Pf: We apply the Jensen's inequality from the concavity of H ,

$$H(p + p' - pp') = H(p \cdot 1 + p'(1 - p)) \leq pH(1) + (1 - p)H(p') = (1 - p)H(p')$$

Information Inequality: Let $p(x), q(x), x \in X$ are probability mass functions then: $D(p||q) \geq 0$ with equality $\iff p(x) = q(x) \forall x$

Pf: Let $A = \{x \in X | p(x) > 0\}$ then $-D(p||q) = -\sum_{x \in A} p(x) \log(\frac{p(x)}{q(x)}) = \sum_{x \in A} p(x) \log(\frac{q(x)}{p(x)}) \leq \log(\sum_{x \in A} p(x) \frac{q(x)}{p(x)}) = \log(\sum_{x \in A} q(x)) \leq \log(\sum_{x \in X} q(x)) = \log(1) = 0 \Rightarrow D(p||q) \geq 0$

Nonnegativity of mutual information: For any two random variables, X, Y : $I(X; Y) \geq 0$ with equality $\iff X, Y$ are independent.

Corollary: $I(X; Y|Z) \geq 0$ with equality $\iff X, Y$ are conditionally independent given Z .

Corollary: $D(p(y|x)||q(y|x)) \geq 0$ with equality $\iff p(y|x) = q(y|x) \forall x, y$ s.t. $p(x) > 0$.

Conditioning reduces entropy: $H(X|Y) \leq H(X)$ with equality $\iff X, Y$ are independent.

Def: Random variables X, Y, Z are said to form a Markov chain in that order $X \rightarrow Y \rightarrow Z$ if the conditional distribution of Z depends only on Y and is conditionally independent of X , i.e., the joint probability mass function can be written as $p(x, y, z) = p(x)p(y|x)p(z|y)$

Data Processing Inequality: If $X \rightarrow Y \rightarrow Z$ then: $I(X; Y) \geq I(X; Z)$ with equality $\iff I(X; Y|Z) = 0$

Corollary: $I(X; Y|Z) \leq I(X; Y)$

3.1.14 Applications of entropy in combinatorics

1. Let $A = (A_1, A_2)$ be a random subset of $\{1, 2\}$ such that each A_i are iid Bernoulli random variables with probability p . Then

$$H(A) = p^2 \log(\frac{1}{p^2}) + 2p(1-p) \log(\frac{1}{p(1-p)}) + (1-p)^2 \log(\frac{1}{(1-p)^2}) = 2p^2 \log(\frac{1}{p}) + 2p(1-p) [\log(\frac{1}{p}) + \log(\frac{1}{(1-p)})] + 2(1-p)^2 \log(\frac{1}{(1-p)}) = 2p \log(\frac{1}{p}) + 2(1-p) \log(\frac{1}{(1-p)}) = 2H(p)$$

In general if $A = (A_1, A_2, \dots, A_n)$ is a random subset of $[n]$ s.t. each A_i are iid Bernoulli random variables with probability p . Then $H(A) = nH(p)$

$$\begin{aligned} H(A) &= \sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} [\log(\frac{1}{p^k}) + \log(\frac{1}{(1-p)^{n-k}})] = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k} \log(\frac{1}{p}) + \\ &\quad \sum_{k=0}^n (n-k) \binom{n}{k} p^k (1-p)^{n-k} \log(\frac{1}{(1-p)}) \\ &= n \log(\frac{1}{p}) \sum_{k=0}^n \binom{n-1}{k-1} p^k (1-p)^{n-k} + n \log(\frac{1}{1-p}) \sum_{k=0}^n \binom{n-1}{k} p^k (1-p)^{n-k} \\ &= np \log(\frac{1}{p}) + n(1-p) \log(\frac{1}{1-p}) = nH(p) \end{aligned}$$

2. Let A and B be two independent samples from the same Bernoulli distribution, i.e. $A = [n]$ with probability p and $A = \emptyset$ with probability $(1 - p)$. Then $H(A) = H(p)$ and $H(A \cup B) = H(2p - p^2)$.

Pf: $Pr[A \cup B = [n]] = Pr[A = [n] \text{ or } B = [n]] = p^2 + 2p(1 - p) = p(2 - p)$

Obviously, $H(A) = H(p)$ and $H(A \cup B) = H(2p - p^2)$. Note that, the graph of $H(p)$,

$H(2p - p^2)$ intersect at $p = \frac{3-\sqrt{5}}{2} \approx 0.38196601125010...$ So $H(A) < H(A \cup B)$ when $p < 0.38196601125010...$

In general, if $A = (A_1, A_2, \dots, A_n)$ be a random subset of $[n]$ s.t. each A_i are iid Bernoulli random variables with probability p . Then $H(A) = nH(p)$ and $H(A \cup B) = nH(2p - p^2)$

Hence it might be possible that 1% density bound could be improved to 38.1966% but no more.

3. Sample $A \subseteq [n]$ in the following manner. First sample A_1 from a Bernoulli distribution with probability p . Then, conditioned on the event that $A_1 = 1$, sample each A_i from iid Bernoulli distributions with probability $q = 0.99$. Otherwise, if $A_1 = 0$ then each $A_i = 0$. To calculate $H(A)$, we apply the chain rule to get $H(A) = H(A_1, A_{>1}) = H(A_1) + H(A_{>1}|A_1)$. The conditional entropy can be computed as $H(A_{>1}|A_1) = Pr[A_1 = 0] \cdot 0 + Pr[A_1 = 1] \cdot H(q)(n-1)$ $H(A) = H(p) + pH(q)(n-1)$. Via a similar calculation we get $H(A \cup B) = H(2p - p^2) + 2p(1-p)H(q)(n-1) + p^2H(2q - q^2)(n-1)$, for n large and p small $H(A \cup B)$ is dominated by $2p(1-p)H(q)(n-1)$. This corresponds to the event that exactly one of A_1, B_1 is equal to 1. So $\frac{H(A \cup B)}{H(A)} \approx 2(1-p)$. the entropy $H(A \cup B|A_1 = B_1 = 1)$ is small relative to $H(A|A_1 = 1)$, i.e., $H(A \cup B|A_1 = B_1 = 1) = H(2q - q^2)(n-1)$ and $H(A|A_1 = 1) = H(q)(n-1)$

4. **Union closed set system:** A set system F is union closed if for all $A, B \in F$ $A \cup B \in F$.
Frankle's Conjecture: (50% density bound) There is an element that appears in at least half the sets of F .

Intuition: Set an injection from the set A of subsets not containing 4 to the set B of subsets containing 4. Like: $\{2, 3\} \rightarrow \{2, 3, 4\}$. Now $|A| = 0.5|P([n])|$. So $|B| \geq 50\%|P([n])|$.

Easier result: There exists $i \in [n] = \{1, 2, \dots, n\}$ which is contained in 1% of the sets in F .

Theorem: Let A and B denote independent samples from a distribution over subsets of $[n]$. Assume that for all $i \in [n]$, $Pr[i \in A] \leq 0.01$. Then $H(A \cup B) \geq 1.26H(A)$.

This will give a contradiction, so that we can conclude $Pr[i \in A] > 0.01$. Theorem $\Rightarrow H(A \cup B) > H(A)$ when $H(A) > 0$. But if we sample A, B independently and uniformly at random from a union-closed family F , then $H(A \cup B) \leq H(A)$. This follows because $H(A \cup B)$ is a distribution over F and the entropy of a distribution over F is maximized when it is the uniform distribution.

Property: $F \subseteq B \Rightarrow \sum_{b \in B} p_b \log(\frac{1}{p_b}) \leq \log(|F|)$

Entropy is maximized when $p_b = \frac{1}{|F|}$ (uniform distribution)

Pf: $\log x$ is concave then applying Jensen's inequality: $\sum_{b \in B} p_b \log(\frac{1}{p_b}) \leq \log(\sum_{b \in B} p_b (\frac{1}{p_b})) = \log(|F|)$

Conjecture 1: If frequency of each element $\leq \frac{1}{|F|}$ then $|F \cup F| = \Omega(|F|^2)$.

$|F \cup F| = \{A \cup B | A, B \in F\}$. We get $|F \cup F| \approx \binom{n}{2p-p^2}$. As $p \rightarrow 0$ $|F \cup F| \rightarrow \Omega(|F|^2)$.

From point 2, we get for $p = \frac{3-\sqrt{5}}{2} \approx 0.38$, $|F \cup F| \approx |F|$.

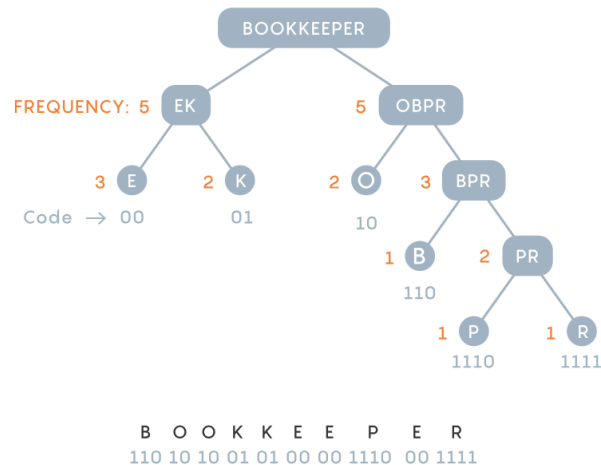
From point 3, we get the ratio of $H(A \cup B)$ and $H(A)$. From there we the value of p so that $H(A \cup B) \leq H(A)$ which satisfies the union-closed family condition.

3.1.15 Data Compression

With more than 9 billion gigabytes of information traveling the internet every day, researchers are constantly looking for new ways to compress data into smaller packages. We have some lossy approach where we intentionally loss some amount of information and transmit it. Google, for instance, recently unveiled a lossy strategy where the sending computer drops details from an image and the receiving computer uses artificial intelligence to guess the missing parts. Even Netflix uses a lossy approach, downgrading video quality whenever the company detects that a user is watching on a low-resolution device.

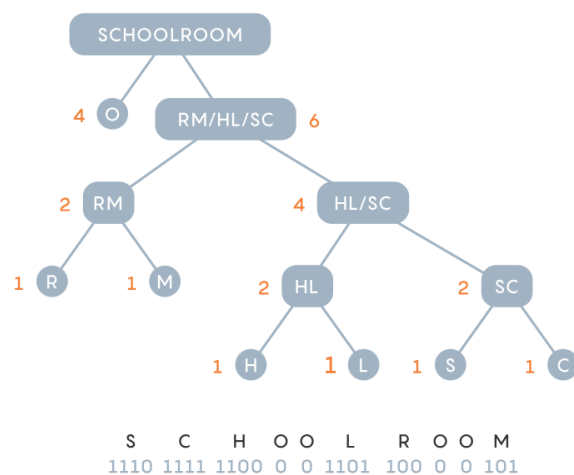
Another approach is we detect more frequent letters and denote it by a symbol. But there is still some problem. In **Morse Code**, recipients would have no way to distinguish the Morse message dash dot-dash-dot dot-dot dash dot ("trite") from dash dot-dash-dot dot-dot-dash dot ("true").

Massachusetts Institute of Technology professor **Robert Fano** solved this part of the problem. To actually determine the codes, Fano built **binary trees**, placing each necessary letter at the end of a visual branch. Each letter’s code was then defined by the path from top to bottom. If the path branched to the left, Fano added a 0; right branches got a 1. The tree structure made it easy for Fano to avoid those undesirable overlaps: Once Fano placed a letter in the tree, that branch would end, meaning no future code could begin the same way.



Consider a message where the Fano approach falters. In “**schoolroom**”, O appears four times, and S/C/H/L/R/M each appear once. Fano’s balancing approach starts by assigning the O and one other letter to the left branch, with the five total uses of those letters balancing out the five appearances of the remaining letters. The resulting message requires 27 bits.

Huffman, by contrast, starts with two of the uncommon letters — say, R and M — and groups them together, treating the pair like a single letter. Huffman again picks the two least common options, matching (say) H with L and so on.

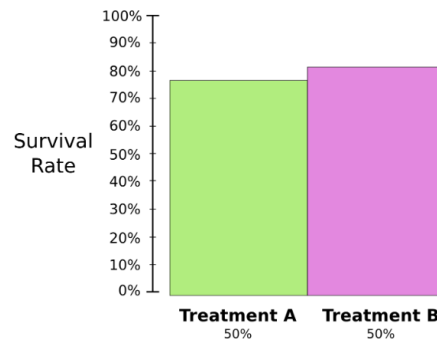


Now schoolroom is encoded in 26 bits. One bit may not sound like much, but even small savings grow enormously when scaled by billions of gigabytes. Indeed, Huffman’s approach has turned out to be so powerful that, today, nearly every lossless compression strategy uses the Huffman insight in whole or in part.

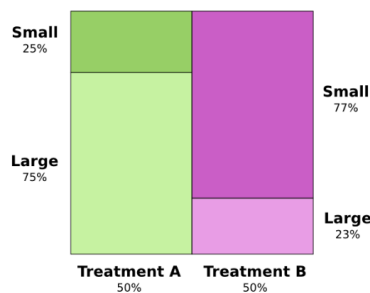
3.1.16 Visualizing Information Theory

Simpson’s Paradox: Simpson’s paradox is an extremely unintuitive statistical situation. Michael Nielsen wrote a lovely essay, *Reinventing Explanation*, which explored different ways to explain it.

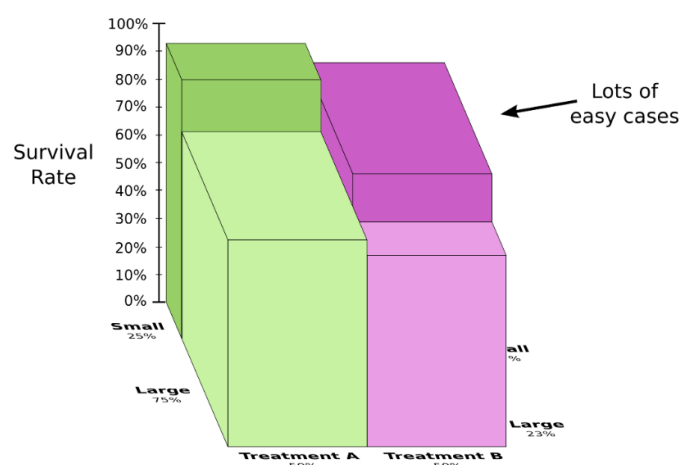
Two treatments for kidney stones are tested. Half the patients are given treatment A while the other half are given treatment B. The patients who received treatment B were more likely to survive than those who received treatment A.



However, patients with small kidney stones were more likely to survive if they took treatment A. Patients with large kidney stones were also more likely to survive if they took treatment A! How can this be?

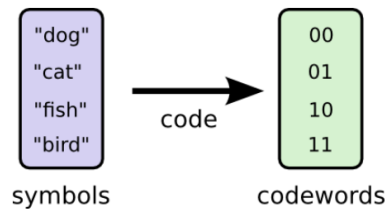


The core of the issue is that the study wasn't properly randomized. The patients who received treatment A were likely to have large kidney stones, while the patients who received treatment B were more likely to have small kidney stones. As it turns out, patients with small kidney stones are much more likely to survive in general. To understand this better, we can combine the two previous diagrams. The result is a 3-dimensional diagram with the survival rate split apart for small and large kidney stones.



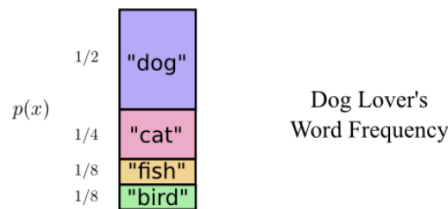
We can now see that in both the small case and the large case, Treatment A beats Treatment B. Treatment B only seemed better because the patients it was applied to were more likely to survive in the first place!

Codes: There are two friends. One of them moved to some far. They decided they only communicate in binary. To communicate, Bob and I have to establish a code, a way of mapping words into sequences of bits.

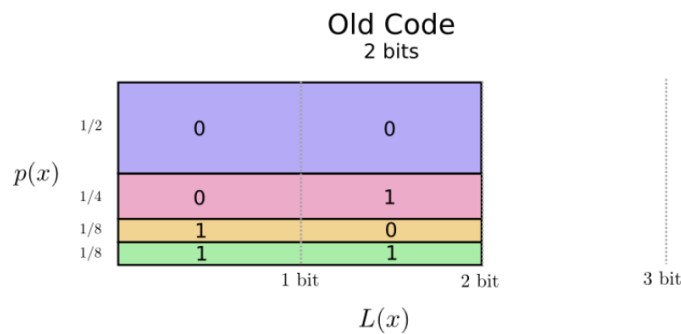


To send a message, Bob replaces each symbol (word) with the corresponding codeword, and then concatenates them together to form the encoded string.

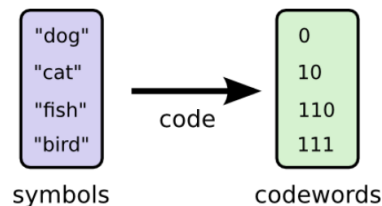
Variable-length Codes: Unfortunately, communication services are expensive. They have to pay \$5 per bit of every message. Then they decided they should investigate whether there is some way they can make their average message length shorter. It turns out, they don't use all words equally often. Here's the graph of word frequency.



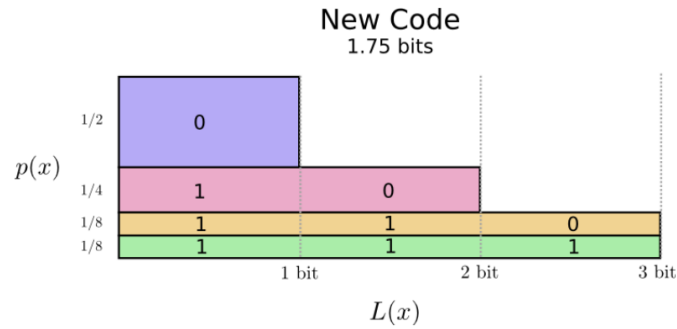
In the following diagram, they use the vertical axis to visualize the probability of each word, $p(x)$ and the horizontal axis to visualize the length of the corresponding codeword, $L(x)$. Notice that the area is the average length of a codeword they send – in this case 2 bits.



To minimize the message length, we'd ideally like all codewords to be short, but they especially want the commonly used ones to be. So the resulting code has shorter codewords for common words (like "dog") and longer codewords for less common words (like "bird").

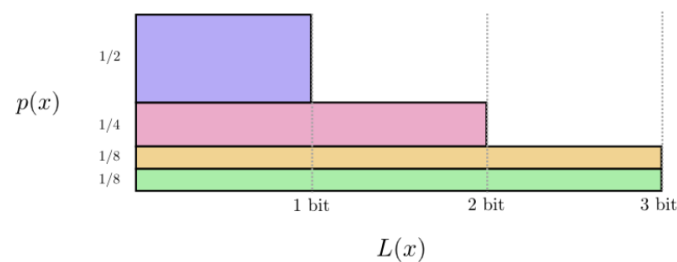


The result is, on net, a smaller amount of area. This corresponds to a smaller expected codeword length. On average, the length of a codeword is now 1.75 bits! It turns out that this code is the best possible code. There is no code which, for this distribution, will give us an average codeword length of less than 1.75 bits.

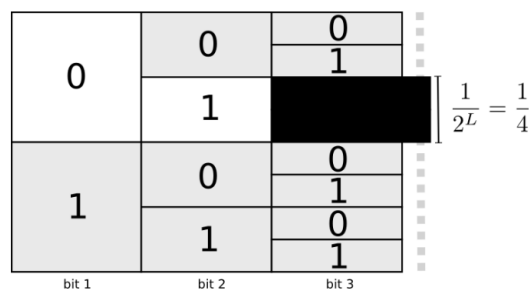


There is simply a fundamental limit. No matter how clever the code, it's impossible to get the average message length to be less. This fundamental limit is called the entropy of the distribution.

Entropy = Optimal Average Length
= Area = 1.75 bits

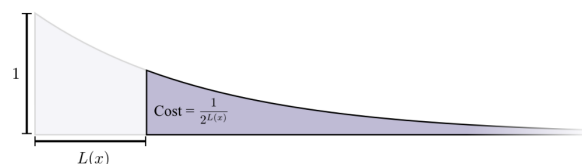


There's a slightly subtle issue one needs to be careful of, when crafting a variable length code. How do we split the encoded string back into the codewords? When all the codewords are the same length, just split the string every couple of steps. But since there are codewords of different lengths, there is only one way to decode the encoded string. For example, imagine that 0 and 01 were both codewords. Then it would be unclear what the first codeword of the encoded string 0100111 is – it could be either! The property we want is that if we see a particular codeword, there shouldn't be some longer version that is also a codeword. Another way of putting this is that no codeword should be the prefix of another codeword. This is called the **prefix property**, and codes that obey it are called **prefix codes**. Every codeword requires a sacrifice from the space of possible codewords.



Since a quarter of all codewords start with 01, we've sacrificed a quarter of all possible codewords. That's the price we pay in exchange for having one codeword that's only 2 bits long! In turn this sacrifice means that all the other codewords need to be a bit longer.

Optimal Encodings: The cost of buying a codeword of length 0 is 1, cost of a codeword of length 1, like "0", is 1/2 because half of possible codewords start with "0". The cost of a codeword of length 2, like "01", is 1/4 because a quarter of all possible codewords start with "01". In general, the cost of codewords decreases exponentially with the length of the codeword.

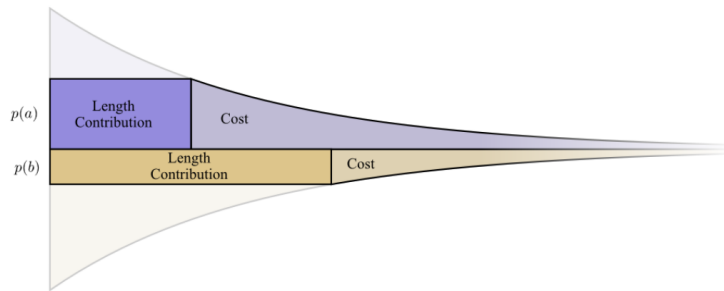


Each codeword makes the average message length longer by its probability times the length of the codeword. For example, if we need to send a codeword that is 4 bits long 50% of the time, our average message length is 2 bits longer than it would be if we weren't sending that codeword. The amount we pay decides the length of the codeword. The length of the codeword controls how much it adds to the average message length. Short codewords reduce the average message length but are expensive, while long codewords increase the average message length but are cheap.

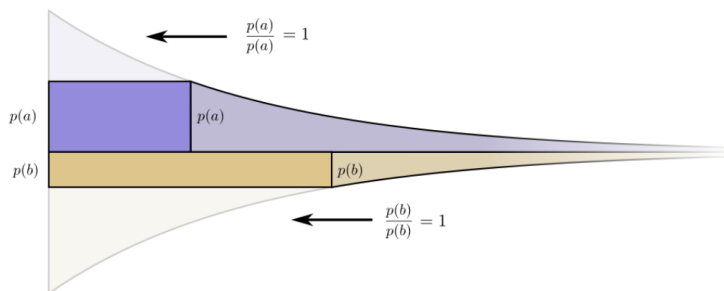


What's the best way to use our limited budget? How much should we spend on the codeword for each event? We want to spend more on frequently used codewords. Distribute the budget in proportion to how common an event is. If one event happens 50% of the time, we spend 50% of our budget buying a short codeword for it. But if an event only happens 1% of the time, we only spend 1% of our budget, because we don't care very much if the codeword is long.

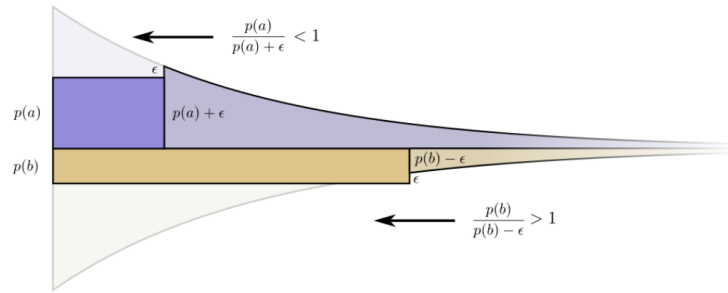
The proof of this is much intuitive. Event a happens $p(a)$ of the time and event b happens $p(b)$ of the time. We distribute our budget in the natural way described above, spending $p(a)$ of our budget on getting a a short codeword, and $p(b)$ on getting b a short codeword.



If we slightly increase the length of the codeword, the message length contribution will increase in proportion to its height at the boundary, while the cost will decrease in proportion to its height at the boundary. So, the cost to make the codeword for a shorter is $p(a)$. The benefit to us of making the codeword for a a bit shorter is $p(a)$. It's interesting that both derivatives are the same. It means that our initial budget has the interesting property that, if you had a bit more to spend, it would be equally good to invest in making any codeword shorter. What we really care about, in the end, is the benefit/cost ratio – that's what decides what we should invest more in. In this case, the ratio is $\frac{p(a)}{p(a)} = 1$ - independent of a . The benefit/cost is always one, so it makes equal sense to invest a bit more in any of them.

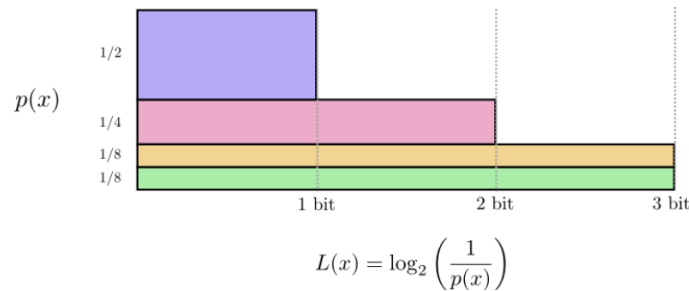


Infinitesimally, it doesn't make sense to change the budget. We'll invest ϵ less in b and invest it in a instead. This makes the codeword for a a bit shorter and the codeword for b a bit longer. Now the cost of buying a shorter codeword for a is $p(a) + \epsilon$ and cost of buying a longer codeword for b is $p(b) - \epsilon$. But the benefits are still the same. This leads the benefit cost ratio for buying a is $\frac{p(a)}{p(a)+\epsilon} < 1$ and benefit cost ratio for buying b is $\frac{p(b)}{p(b)-\epsilon} > 1$.



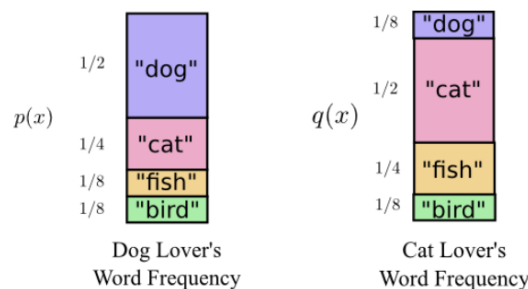
The prices are no longer balanced. b is better deal than a . The investors scream: “Buy b ! Sell a !” We do this, and end back at our original budget plan. All budgets can be improved by shifting towards our original plan. The original budget – investing in each codeword in proportion to how often we use it – wasn’t just the natural thing to do, it was the optimal thing to do.

Calculating Entropy: The cost of a message of length L is $Cost = \frac{1}{2^L}$. We can get the length from $\log_2(\frac{1}{Cost})$. Since we spend $p(x)$ for x , the length is $\log_2(\frac{1}{p(x)})$.

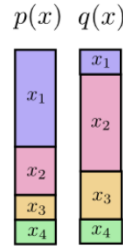


So, Entropy = $H(p) = \sum_x p(x) \log_2(\frac{1}{p(x)})$. No matter what we do, on average we need to send at least that number of bits if we want to communicate which event occurred. If we knew for sure what was going to happen, we wouldn’t have to send a message at all! If there’s two things that could happen with 50% probability, we only need to send 1 bit. But if there’s 64 different things that could happen with equal probability, we’d have to send 6 bits. The more uncertain the outcome, the more we learn, on average, when we find out what happened.

Cross-Entropy: Let 2 friends use the same words with different frequencies. Initially, Alice sent messages using Bob’s code. Unfortunately, her messages were longer than they needed to be. Bob’s code was optimized to his probability distribution. Alice has a different probability distribution, and the code is suboptimal for it. While the average length of a codeword when Bob uses his own code is 1.75 bits, when Alice uses his code it’s 2.25. It would be worse if the two weren’t so similar!



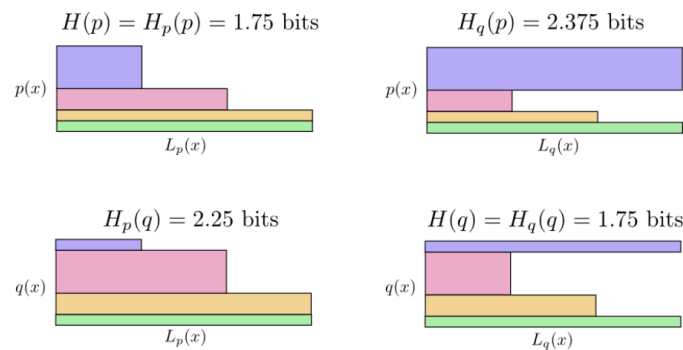
The average length of communicating an event from one distribution with the optimal code for another distribution – is called the cross-entropy. Formally, we can define cross-entropy as: $H_p(q) = \sum_x q(x) \log_2(\frac{1}{p(x)})$. In this case, it’s the cross-entropy of Alice the cat-lovers word frequency with respect to the Bob the dog-lovers word frequency.



Cross-Entropy: $H_p(q)$

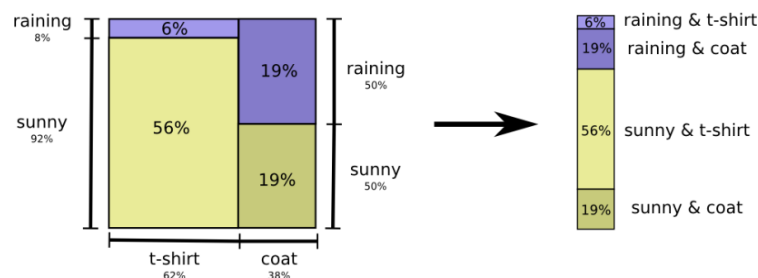
Average Length
of message from $q(x)$
using code for $p(x)$.

So, now we have four possibilities: (i) Bob using his own code ($H(p) = 1.75$ bits), (ii) Alice using Bob's code ($H_p(q) = 2.25$ bits), (iii) Alice using her own code ($H(q) = 1.75$ bits), (iv) Bob using Alice's code ($H_q(p) = 2.375$ bits). In the following diagram, each subplot represents one of these 4 possibilities. Each subplot visualizes average message length. We can see that $H_p(q) \neq H_q(p)$. $H_q(p)$ is large because there is an event (blue) which is very common under p but gets a long code because it is very uncommon under q . On the other hand, common events under q are less common under p , but the difference is less drastic, so $H_p(q)$ isn't as high.

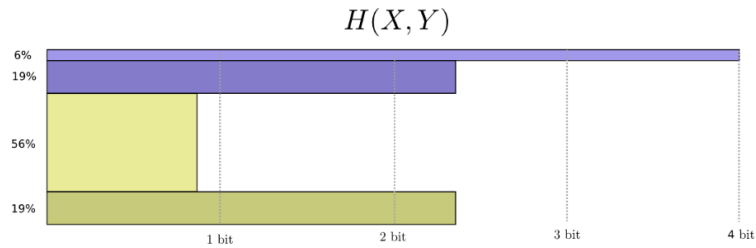


Cross-entropy isn't symmetric. The more different the distributions p and q are, the more the cross-entropy of p wrt q will be bigger than the entropy of p . Similarly, the more different p is from q the more the cross-entropy of q wrt p will be bigger than the entropy of q . The difference between the entropy and the cross-entropy is how much longer our messages are because we used a code optimized for a different distribution. If the distributions are the same, this difference will be zero. As the difference grows, it will get bigger. We call this difference the **Kullback–Leibler divergence**. The KL divergence of p wrt $q = D_q(p) = H_q(p) - H(p)$. The really neat thing about KL divergence is that it's like a distance between two distributions. It measures how different they are! Cross-Entropy and KL divergence are incredibly useful in **machine learning**. Often, we want one distribution to be close to another. For example, we might want a predicted distribution to be close to the ground truth. KL divergence gives us a natural way to do this, and so it shows up everywhere.

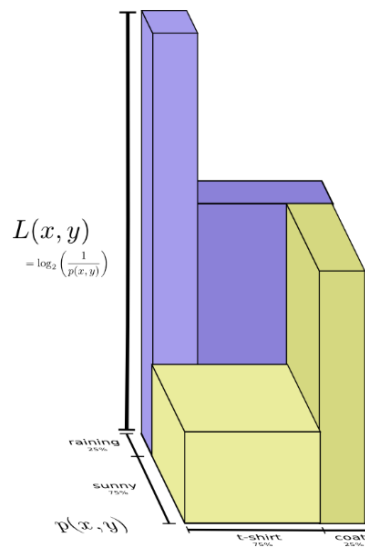
Entropy & Multiple Variables:



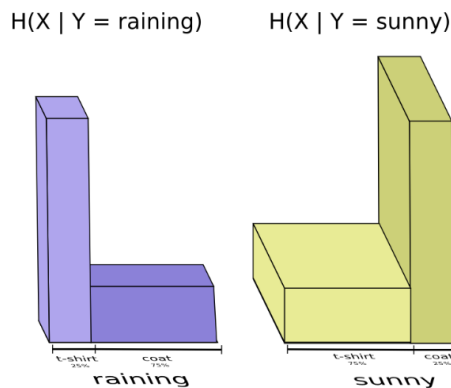
Let the following is the joint distribution of the weather and dress. Now we can figure out the optimal codewords for events of these probabilities and compute the average message length:



We call this the joint entropy of X and Y : $H(X, Y) = \sum_{x,y} p(x, y) \log_2 \left(\frac{1}{p(x, y)} \right)$. A slightly nicer way to think about this is to avoid flattening the distribution, and just think of the code lengths as a third dimension. Now the entropy is the volume!

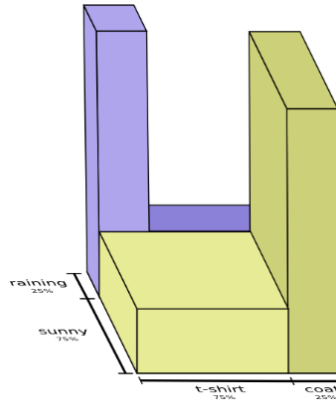


But suppose my mom already knows the weather. She can check it on the news. Now how much information do I need to provide? Let's consider the case where it's raining and where it's sunny separately.



In both cases, I don't need to send very much information on average, because the weather gives me a good guess at what the right answer will be. When it's sunny, I can use a special sunny-optimized code, and when it's raining I can use a raining optimized code. In both cases, I send less information than if I used a generic code for both. To get the average amount of information I need to send my mother, I just put these two cases together...

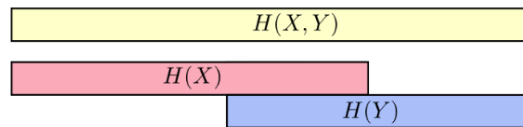
$$H(X|Y) = 0.81 \text{ bits}$$



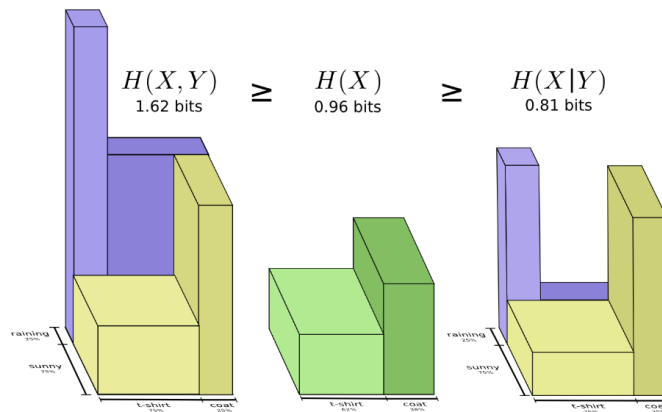
We call this the conditional entropy. If we formalize it into an equation, we get:

$$H(X|Y) = \sum_y p(y) \sum_x p(x|y) \log_2\left(\frac{1}{p(x|y)}\right) = \sum_{x,y} p(x,y) \log_2\left(\frac{1}{p(x|y)}\right)$$

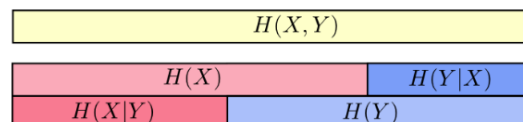
Mutual Information: We observed that knowing one variable can mean that communicating another variable requires less information. One nice way to think about this is to imagine amounts of information as bars. These bars overlap if there's shared information between them. For example, some of the information in X and Y is shared between them, so $H(X)$ and $H(Y)$ are overlapping bars. And since $H(X,Y)$ is the information in both, it's the union of the bars $H(X)$ and $H(Y)$.



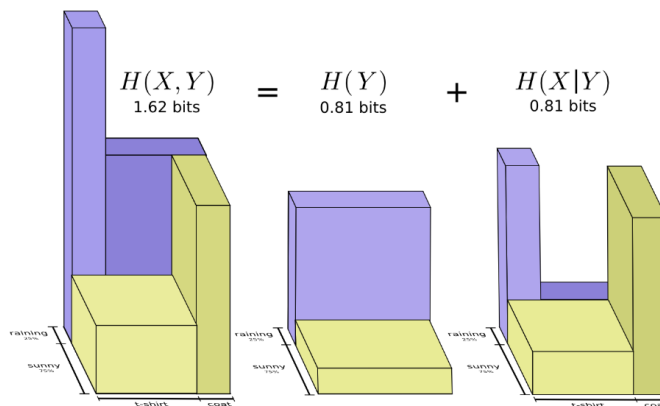
We previously noted it takes more information to communicate both X and Y than it takes to just communicate X or Y . But if you already know Y then it takes less information to communicate X .



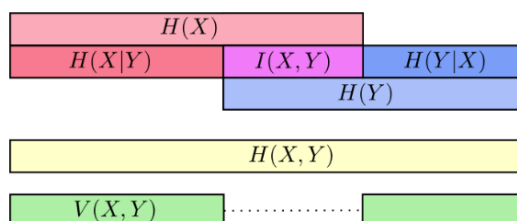
It's very simple when we think about it from the bar perspective. $H(X|Y)$ is the information we need to send to communicate X who already knows Y , the information in X , which isn't in Y . Visually, that means $H(X|Y)$ is the part of $H(X)$ bar which doesn't overlap with $H(Y)$. So we have: $H(X,Y) \geq H(X) \geq H(X|Y)$.



Another identity is that: $H(X, Y) = H(Y) + H(X|Y)$. That is, the information in X and Y is the information in Y plus the information in X which is not in Y .



We have $H(X), H(Y), H(X, Y), H(X|Y)$. We saw the shared part of the bars which is the intersection of $H(X)$ and $H(Y)$. We call this **Mutual Information** $= I(X, Y) = H(X) + H(Y) - H(X, Y)$. Closely related to the mutual information is the variation of information. The **variation of information** is the information which isn't shared between the variables. We can define it like so: $V(X, Y) = H(X, Y) - I(X, Y)$. We can bring this all together into a single diagram relating all these different kinds of information:

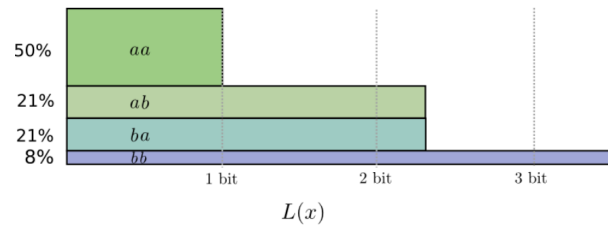


Variation of information is interesting because it gives us a metric, a notion of distance, between different variables. The variation of information between two variables is zero if knowing the value of one tells you the value of the other and increases as they become more independent. KL divergence gives us a distance between two distributions over the same variable or set of variables. In contrast, variation of information gives us distance between two jointly distributed variables. KL divergence is between distributions.

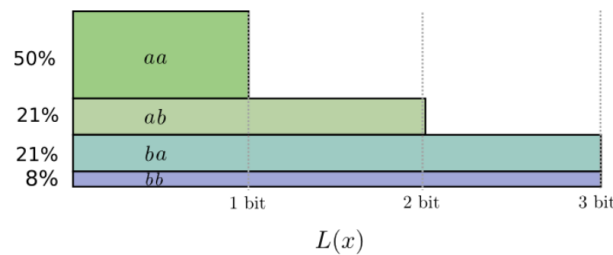
Fractional Bits: A very un-intuitive thing about information theory is that we can have fractional numbers of bits. Often, we're interested in the average length of a message rather than any particular message length. If half the time one sends a single bit, and half the time one sends two bits, on average one sends one and a half bits. Often, the optimal lengths of codewords are fractional. Let's consider a probability distribution where one event a happens 71% of the time and another event b occurs 29% of the time. The optimal code would use 0.5 bits to represent a and 1.7 bits to represent b . Well, if we want to send a single one of these codewords, it simply isn't possible. We're forced to round to a whole number of bits, and send on average 1 bit. But if we're sending multiple messages at once, it turns out that we can do better. Let's consider communicating two events from this distribution. If we sent them independently, we'd need to send 2 bits.

aa 50% 1 bit	ab 21% 2.3 bits
ba 21% 2.3 bits	bb 8% 3.5 bits

Half the time, we need to communicate aa , 21% of the time we need to send ab or ba and 8% of the time we need to communicate bb . Again, the ideal code involves fractional numbers of bits.



If we round the codeword lengths, we'll get something like this:



This codes give us an average message length of 1.8 bits. That's less than the 2 bits when we send them independently. Another way of thinking of this is that we're sending 0.9 bits on average for each event. If we were to send more events at once, it would become smaller still. As $n \rightarrow \infty$, the overhead due to rounding our code would vanish, and the number of bits per codeword would approach the entropy. Further, notice that the ideal codeword length for a was 0.5 bits, and the ideal codeword length for aa was 1 bit. Ideal codeword lengths add, even when they're fractional! So, if we communicate a lot of events at once, the lengths will add. (Huffman coding, which is basically the kind of code we've sketched out here, doesn't handle fractional bits very gracefully. **Arithmetic coding** is a bit different, but elegantly handles fractional bits to be asymptotically optimal.)

3.2 Introduction to Algebraic Coding Theory

3.2.1 Goal of Coding Theory

Let there is a message X with length k . Let we define an encoding where message X becomes codeword C with length n . Obviously $n > k$ as we are adding some redundancy. Now we transmit the codeword C , some parts of C is corrupted and the receiver gets codeword $\tilde{C}$. Our goal is to find such an encoding s.t. we can recover X or some information about X from $\tilde{C}$.

Example 1: (Communication) 2 friends Alice and Bob want to communicate. There is a noisy channel between them. In general we can replace them with a cellphone and cell-tower or satellite and satellite dish. Let Alice wants to send a message $X \in \{0, 1\}^k$. The codeword for X is $C \in \{0, 1\}^n$ which is transmitted through the channel. Bob receives $\tilde{C}$. Bob needs to recover X from $\tilde{C}$.

Example 2: (Storage) Let we want to store file $X \in \{0, 1\}^k$ in some storage. As there are noisy channel the storing is not perfect. But we can't lose our data. So first we encode the file into codeword $C \in \{0, 1\}^n$ and store C in storage, let hard-disc. Let the hard-disc receives $\tilde{C}$. How can we recover file X .

We need to care about - (i) handling corruption, (ii) recovering information about X , (iii) minimizing overhead ($\frac{k}{n}$ should be large, closed to 1), (iv) doing these efficiently.

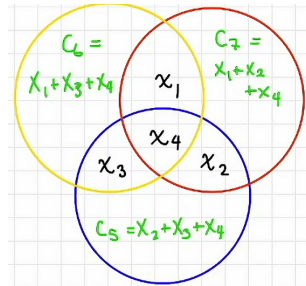
3.2.2 Definitions

Def 1: Given a code $C \subseteq \Sigma^n$, its dimension is given by $k = \log_q |C|$ where $q = |\Sigma|$.

Def 2: A code C of block-length n over an alphabet Σ is $C \subseteq \Sigma^n$. The elements $c \in C$ are codewords.

Def 3: An equivalent description of the code $C \subseteq \Sigma^n$ is an injective mapping $E : [|C|] \rightarrow \Sigma^n$ called an **Encoding function** for C and a mapping $D : \Sigma^n \rightarrow [|C|]$ is called a **Decoding function** for C .

Example 1: $C = \{(0, 0, 0, 0), (0, 0, 1, 1), (0, 1, 0, 1), (0, 1, 1, 0), (1, 0, 0, 1), (1, 0, 1, 0), (1, 1, 0, 0), (1, 1, 1, 1)\}$ is a code of length 4 over $\Sigma = \{0, 1\}$. Consider the encoding map $ENC : \{0, 1\}^3 \rightarrow \{0, 1\}^4$ where $ENC(x_1, x_2, x_3) = (x_1, x_2, x_3, (x_1 + x_2 + x_3) \bmod 2)$. We can see $C = Im(ENC)$ i.e., C has all vectors of the form $(x_1, x_2, x_3, (x_1 + x_2 + x_3) \bmod 2)$. This example can correct one **ERASURE**. e.g., let $\tilde{C} = (0, -, 0, 1)$. Then 2nd term must be 1, otherwise it doesn't satisfy 4th term $= (x_1 + x_2 + x_3) \bmod 2$. So we have recovered $c \in C$. We know which bit is erased but don't know the value. If $\tilde{C} = (0, 0, 0, 1)$ then this is not possible. If a bit is wrong then that's called an error. This example can detect one error. Then we can correct $\tilde{C}$. Consider $ENC : \{0, 1\}^4 \rightarrow \{0, 1\}^7$ where $ENC(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, x_4, (x_2 + x_3 + x_4) \bmod 2, (x_1 + x_3 + x_4) \bmod 2, (x_1 + x_2 + x_4) \bmod 2)$. $C = Im(ENC)$.



We can visualize C considering 3 sets. The sum of each circle's components $= 0 \bmod 2$. From that we can detect one error.

Example 2: Let $\Sigma = \{0, 1\}$. Code C has 4 bit vectors. Given a message $(x_1, x_2, x_3, x_4) \in \Sigma^4$ it's corresponding codeword is $C_{\oplus}(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, x_4, x_1 \oplus x_2 \oplus x_3 \oplus x_4)$. $\oplus$ denotes XOR operation. This code is called **Parity code**. In other words, the parity code appends the parity of the message bits (or takes the remainder of the sum of the message bits when divided by 2) at the end of the message. Note that such a code uses the minimum amount of non-zero redundancy.

Example 3: Let C has 4 bits and each bit is used 3 times. This code is called **Repetition code** and denoted as $C_{3,rep}$. This is a simple mapping. But it has lots of redundancy.

Proposition: $C_{3,rep}$ is a 1-error correcting code.

For example, if both of the errors happen in the same block and a block in the received word is 010, then the original block in the codeword could have been either 111 or 000. Therefore in this

case, no decoder can successfully recover the transmitted message.

Def 4: The **Hamming distance** between 2 vectors $x, y \in \Sigma^n$ is $\Delta(x, y) = \sum_{i=1}^n 1\{x_i \neq y_i\}$ where 1 is indicator. The **Relative Hamming distance** between 2 vectors $x, y \in \Sigma^n$ is $\delta(x, y) = \frac{\Delta(x, y)}{n}$

Def 5: The **Minimum distance** of a code $C \subseteq \Sigma^n$ is $\min_{c \neq c' \in C} \Delta(c, c')$

Thm 1: A code of Minimum distance d can : (i) CORRECT $\leq d - 1$ ERASURES, (ii) DETECT $\leq d - 1$ ERRORS, (iii) CORRECT $\leq \lfloor \frac{d-1}{2} \rfloor$ ERRORS.

Proof can be seen from an algorithm where we take Hamming balls of radius $\leq \lfloor \frac{d-1}{2} \rfloor$ around c and c' and if $\tilde{c}$ is corrupted codeword then return $c \in C$ s.t. $\Delta(c, \tilde{c})$ is minimized. If $\tilde{c} \in C$ then "NO ERROR", else "ERROR".

Algorithm (Naive Maximum Likelihood Decoder): INPUT: Received word $y \in \Sigma^n$ and OUTPUT: $D_{MLD}(y) = \arg \min_{c \in C} \Delta(c, y)$

```

1: Pick an arbitrary  $c \in C$  and assign  $z \leftarrow c$ 
2: FOR every  $c' \in C$  such that  $c \neq c'$  DO
3: IF  $\Delta(c', y) < \Delta(z, y)$  THEN
4:  $z \leftarrow c'$ 
5: Return  $z$ .

```

Def 6: The **Message length** of a code C over an alphabet Σ is $k = \log_{|\Sigma|} |C|$. As we have ENC : message over Σ of length $k \rightarrow$ codeword $c \in C$, so $|\Sigma|^k = |C| \Rightarrow k = \log_{|\Sigma|} |C|$

Def 7: The **Rate** of a code $C \subseteq \Sigma^n$ is $R = \frac{\log_{|\Sigma|} |C|}{n} = \frac{\text{message length}}{\text{block length}} \in [0, 1]$. If R is close to 1 then it's good, not much overhead, if R is close to 0 then it's bad, lots of overhead.

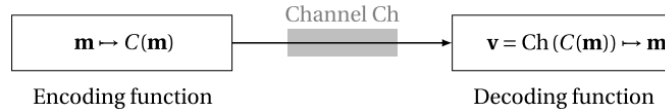
Def 8: A code with distance d , block length n , message length k over an alphabet Σ is called $(n, k, d)_{|\Sigma|}$ code.

Def 9: A **Hamming ball** in Σ^n of radius e about $x \in \Sigma^n$ is $B_{\Sigma^n}(x, e) = \{y \in \Sigma^n | \Delta(x, y) \leq e\}$. The **Volume** of $B_{\Sigma^n}(x, e)$ is $Vol_{|\Sigma|}(e, n) = |B_{\Sigma^n}(0, e)|$. Also, $Vol_q(e, n) = 1 + \binom{n}{1}(q-1) + \binom{n}{2}(q-1)^2 + \dots + \binom{n}{e}(q-1)^e$.

Def 10: For a vector $x \in \Sigma^n$ where $0 \in \Sigma$, $wt(x) = \#$ nonzero entries of x .

Def 11: An n -symbol t -error channel over the alphabet Σ is a function $Ch : \Sigma^n \rightarrow \Sigma^n$ that satisfies $\Delta(v, Ch(v)) \leq t \forall v \in \Sigma^n$.

Def 12: $C \subseteq \Sigma^n$ is said to be a t -error correcting code if there exists a decoding function D such that for every message $m \in [|C|]$ every t -error channel Ch we have $D(Ch(C(m))) = m$. Thus a t -error-correcting code is one where there is a decoding function that corrects any pattern of t errors.



Def 13: $C \subseteq \Sigma^n$ is said to be a t -error detecting code if there exists a detecting procedure D such that for every message $m \in [|C|]$ every received vector $v \in \Sigma^n$ we have $\Delta(C(m), v) \leq t$, it holds that D outputs a 1 if $v = C(m)$ and 0 otherwise. So t -error correcting code is also a t -error detecting code.

Def 14: An n -symbol t -Erasure Channel over the alphabet Σ is a function $Ch : \Sigma^n \rightarrow (\Sigma \cup \{a\})^n$ that satisfies $\Delta(v, Ch(v)) \leq t$ for every $v \in \Sigma^n$ (where both arguments to $\Delta(\cdot, \cdot)$ are viewed as elements of $(\Sigma \cup \{a\})^n$ and for every $i \in [n]$ such that $v_i \neq Ch(v)_i$ we have $Ch(v)_i = a$. A coordinate i such that $Ch(v)_i = a$ is called an Erasure.

Def 15: $C \subseteq \Sigma^n$ is said to be a t -erasure correcting code if there exists a decoding function D such that for every message $m \in [|C|]$ every t -erasure channel Ch we have $D(Ch(C(m))) = m$.

3.2.3 Channel Noise

Any error pattern can occur during transmission as long as the total number of errors is bounded. We will frequently refer to Hamming's model as the Adversarial Noise Model. So for **example**, if the error pattern is $(1, 0, 1, 0, 0, 0)$ and we consider the alphabet to be $\{0, 1\}$, then the pattern has two errors. However, if our alphabet is $\{0, 1\}^3$ (i.e. we think of the vector above as $((1, 0, 1), (0, 0, 0))$, with $(0, 0, 0)$ corresponding to the zero element in $\{0, 1\}^3$, then the pattern has only one error. Thus, by increasing the alphabet size we can also change the adversarial noise model. For **example**, let's consider the error model: No more than 1 error can happen in any contiguous three-bit block. Then no more than four errors can occur when a codeword in $C_{3,rep}$ is transmitted.

Note that the decoding function that takes the majority vote of each block can successfully recover the transmitted codeword for any error pattern, while in the worst-case noise model it could only correct at most one error. A popular alternate noise model is to model the channel as a stochastic process. A concrete **example**, let us briefly mention the binary symmetric channel with crossover probability $0 \leq p < 1$, denoted by BSC_p , which was first studied by Shannon. In this model, when a (binary) codeword is transferred through the channel, every bit flips independently with probability p . Note that the two noise models proposed by Hamming and Shannon are in some sense two extremes: Hamming's model assumes no knowledge about the channel (except that a bound on the total number of errors is known while Shannon's noise model assumes complete knowledge about how noise is produced. In real life, the situation often is somewhere in between. For real life applications, modeling the noise model correctly is an extremely important task, as we can tailor our codes to the noise model at hand. Better is that we have to consider the worst-case noise model. Intuitively, if one can communicate over the worst-case noise model, then one could use the same code to communicate over nearly every other noise model with the same amount of noise.

Algorithm (Error Detector for Parity Code): INPUT: Received word $y = (y_1, y_2, y_3, y_4, y_5)$ and OUTPUT: 1 if $y \in C_{\oplus}$ and 0 otherwise.

1: $b \leftarrow y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5$

2: Return $1 \oplus b$. (If there is no error, then $b = 0$ and hence we need to "flip" the bit for the answer)

Proposition: $C_{\oplus}$ can detect an odd number of errors.

3.2.4 Hamming Code

Remember the previous example where we had an encoding map $ENC : \Sigma^4 \rightarrow \Sigma^7$ and $ENC(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, x_4, (x_2 + x_3 + x_4) \bmod 2, (x_1 + x_3 + x_4) \bmod 2, (x_1 + x_2 + x_4) \bmod 2)$. This example is called **Hamming Code of length 7**. Let $C = Im(ENC)$. We can

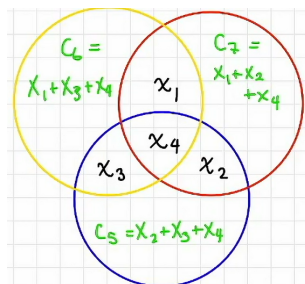
write the map in matrix multiplication: $ENC(\bar{x}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \bmod 2 = G \cdot \bar{x} \bmod 2$

where G is this matrix. G is called a **Generator matrix**.

Linear algebraic observations: (i) C is closed under addition, i.e., for any $c_1, c_2 \in C$ $c_1 + c_2 \in C$. because, $c_1 + c_2 = G \cdot x_1 + G \cdot x_2 = G \cdot (x_1 + x_2) \in C$.

(ii) C is a linear subspace of $\{0, 1\}^7$ of dimension 4, as C is closed under addition and scalar multiplication in mod 2 and $C = \text{Column-span}(G)$ and dimension of G is 4, because we can see the first 4 rows of G are lin independent and forms identity matrix.

(iii) The distance of C is the same as the minimum weight of any nonzero $c \in C$, because: $\Delta(c_1, c_2) = \Delta(Gx_1, Gx_2) = \Delta(G(x_1 - x_2), 0) = \Delta(c', 0) = wt(c')$, where $c' = G(x_1 - x_2) \in C$, so minimum Hamming distance of code C will be minimum weight of any nonzero $c' \in C$.



Another way to write: we take $x_1, x_2, x_3, x_4, c_5, c_6, c_7$ in matrix multiplication:

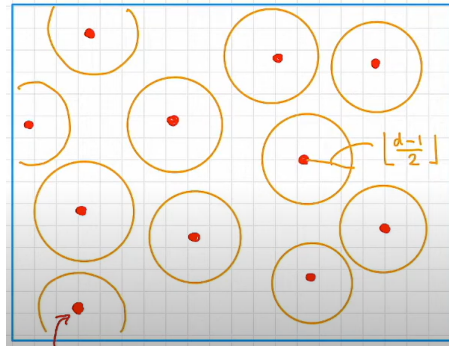
$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ c_5 \\ c_6 \\ c_7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ mod } 2. \text{ Call this matrix } H. \text{ So for any codeword } c \in C$$

$H \cdot c = 0 \Rightarrow C \subseteq \text{Ker}(H)$. **Claim:** $\text{Ker}(H) = C$. $\dim(\text{Ker}(H)) = 7 - \dim(\text{Rank}(H)) = 7 - 3 = 4$ because we can see each rows of H is lin independent as the last three columns forms identity matrix. Also, $\dim(C) = 4 = \dim(\text{Ker}(H)) \Rightarrow C = \text{Ker}(H)$. H is called **Parity-Check matrix** for C . C has distance 3 so it suffices to show minimum weight of nonzero element is 3. Let C has minimum weight 1 or 2. Then there's some nontrivial linear combination of ≤ 2 of the columns of H is 0 mod 2. But any pair of columns of H are lin independent, so contradiction! So $\text{distance}(C) \geq 3$. Now consider the codeword (0101010) has weight 3 $\Rightarrow \text{distance}(C) \leq 3 \Rightarrow \text{distance}(C) = 3$.

Efficient decoding algorithm: $\tilde{c} = 0111010$, find $c \in C$ s.t. $\Delta(c, \tilde{c}) \leq 1$. $H \cdot \tilde{c} = (1, 1, 0)^t$ and $\tilde{c} = c + z$ where $z = \text{weight } 1 \text{ error vector mod } 2$. So $H\tilde{c} = Hc + Hz = Hz \text{ mod } 2$, as $c \in C$ $Hc = 0$. Then $Hz = (1, 1, 0)^t \text{ mod } 2 \Rightarrow z$ must be $(0, 0, 1, 0, 0, 0, 0)^t \Rightarrow c = \tilde{c} - z = (0, 1, 0, 1, 0, 1, 0)^t$.

3.2.5 Hamming Bound

Let we have some points in a plane with minimum distance d . Now we draw hamming balls around each points of radius $\lfloor \frac{d-1}{2} \rfloor$. Then each 2 balls are disjoint.



Let $C \subseteq \Sigma^n$ be a code with distance d and message length k . Let $q = |\Sigma|$, then total volume of disjoint Hamming balls = $|C| \cdot \text{Vol}_q(\lfloor \frac{d-1}{2} \rfloor, n) \leq q^n = \text{volume of } \Sigma^n \Rightarrow \log_q(|C|) + \log_q(\text{Vol}_q(\lfloor \frac{d-1}{2} \rfloor, n)) \leq n \Rightarrow \text{Rate} = \frac{\log_q(|C|)}{n} \leq 1 - \frac{\log_q(\text{Vol}_q(\lfloor \frac{d-1}{2} \rfloor, n))}{n}$. This relation is called **Hamming Bound**.

Theorem: Every binary code with block length n , dimension k , distance $d = 3$ satisfies $k \leq n - \log_2(n + 1)$.

Pf: Given any two codewords $c_1 \neq c_2 \in C$ $B(c_1, 1) \cap B(c_2, 1) = \emptyset$. Note that $\forall x \in \{0, 1\}^n$ $|B(x, 1)| = n + 1$. Now consider the union of all Hamming balls centered around some codeword. Obviously, their union is a subset of $\{0, 1\}^n$, i.e., $|\bigcup_{c \in C} B(c, 1)| \leq 2^n$. Now $|\bigcup_{c \in C} B(c, 1)| = \sum_{c \in C} |B(c, 1)| = \sum_{c \in C} (n + 1) = 2^k(n + 1)$ where C has dimension k . We get, $2^k(n + 1) \leq 2^n \Rightarrow 2^k \leq \frac{2^n}{n+1} \Rightarrow k \leq n - \log_2(n + 1)$.

3.2.6 Generalized Hamming Bound

Def: A code $C \in \Sigma^n$ with dimension k and distance d will be called a $(n, k, d)_\Sigma$ code.

Theorem: For every $(n, k, d)_{|\Sigma|}$ code: $k \leq n - \log_{|\Sigma|}(\sum_{i=0}^{\lfloor \frac{d-1}{2} \rfloor} \binom{n}{i} (q-1)^i)$. (here, $q = |\Sigma|$)

Pf: Let $e = \lfloor \frac{d-1}{2} \rfloor$. Given 2 codewords $c_1 \neq c_2 \in C$ $B(c_1, e) \cap B(c_2, e) = \emptyset$. Assume, $y \in B(c_1, e) \cap B(c_2, e)$ then $\Delta(y, c_1) \leq e$ and $\Delta(y, c_2) \leq e$. By triangle inequality $\Delta(c_1, c_2) \leq 2e \leq d-1$, which is a contradiction.

We claim that $\forall x \in [q]^n$: $|B(x, e)| = \sum_{i=0}^e \binom{n}{i} (q-1)^i$. Any vector in $B(x, e)$ must differ from x in exactly $0 \leq i \leq e$ positions. In the summation $\binom{n}{i}$ is the number of ways of choosing the differing i positions and in each such position, a vector can differ from x in $q-1$ ways. Now consider the union

of all Hamming balls centered around some codeword. Obviously, their union is a subset of $[q]^n$. In other words, $|\bigcup_{c \in C} B(c, e)| \leq q^n$ and $|\bigcup_{c \in C} B(c, e)| = \sum_{c \in C} |B(c, e)| = q^k \sum_{i=0}^e \binom{n}{i} (q-1)^i$. So combining we get $k \leq n - \log_q \left(\sum_{i=0}^{\lfloor \frac{d-1}{2} \rfloor} \binom{n}{i} (q-1)^i \right)$.

In particular, any code of distance d can have rate R at most $1 - \frac{\log_q \left(\sum_{i=0}^{\lfloor \frac{d-1}{2} \rfloor} \binom{n}{i} (q-1)^i \right)}{n}$. This bound is called the **Hamming Bound** and the codes which meet Hamming bound is called **Perfect Codes**.

3.3 Probabilistic Method

3.3.1 Basic Probabilistic Method & Examples

The basic Probabilistic Method can be described as follows: in order to prove the existence of a combinatorial structure with certain properties, we construct an appropriate probability space and show that a randomly chosen element in this space has the desired properties with positive probability. This method is initiated by Paul Erdos that's why it is called "**The Erdos Method**".

Using probabilistic method we can get lower bound for **Ramsey number** $R(k, l)$. The Ramsey-number $R(k, l)$ is the smallest integer n such that in any 2-coloring of the edges of a complete graph on n vertices K_n by red or blue either there is red K_k or blue K_l . Ramsey showed that $R(k, l)$ is finite for finite k, l .

Proposition 1: If $\binom{n}{k} 2^{1-\binom{k}{2}} < 1$ then $R(k, k) > n$. Thus $R(k, k) > \lfloor 2^{\frac{k}{2}} \rfloor \forall k \geq 3$.

Proof: Consider 2-coloring of the edges of K_n . Prob of each edge is red or blue is $\frac{1}{2}$. First we choose k vertices from n vertices in $\binom{n}{k}$ ways. Then the subgraph of those k vertices has $\binom{k}{2}$ edges. So $P(\text{monochromatic subgraph}) = \frac{2}{2^{\binom{k}{2}}}$ as the subgraph will be red or blue colored.

So $P(\text{monochromatic subgraph}) = \binom{n}{k} 2^{1-\binom{k}{2}} < 1$. Thus with positive prob, there should exists red or blue monochromatic subgraph K_k i.e., $R(k, k) > n$. For $k \geq 3$, we take $n = \lfloor 2^{\frac{k}{2}} \rfloor$ then $\binom{n}{k} 2^{1-\binom{k}{2}} < \frac{2^{1+\frac{k}{2}}}{k!} \cdot \frac{n^k}{2^{\frac{k^2}{2}}} < 1$ hence $R(k, k) > \lfloor 2^{\frac{k}{2}} \rfloor$.

Def 1: A tournament graph has **property** S_k if for every set of k players there is one player who beats them all.

Thm 1: If $\binom{n}{k} (1 - 2^{-k})^{n-k} < 1$ then there is a tournament on n vertices that has property S_k .

Proof: Consider a random tournament on set of n vertices. Now fix a subset K with k vertices of V , let A_k be the event that there is no vertex who beats all members of K . Then $P(A_k) = (1 - 2^{-k})^{n-k}$. Now $P(\sum_{\text{all } k \text{ subsets}} A_k) \leq \sum_{\text{all } k \text{ subsets}} P(A_k) = \binom{n}{k} (1 - 2^{-k})^{n-k} < 1$. So with positive prob no event A_k occurs, there is a tournament graph which has property S_k . We can get approximation of this expression, i.e., $\binom{n}{k} < (\frac{en}{k})^k$ and $(1 - 2^{-k})^{n-k} < e^{-\frac{n-k}{2^k}}$. If $f(k)$ = minimum possible number of vertices of a tournament that has the property S_k then $f(k) \leq k^2 \cdot 2^k \cdot (\ln 2)(1 + o(1))$.

Thm 2: $E(\sum_i A_i) = \sum_i E(A_i)$ (linearity of expectation)

Def 2: A dominating set of an undirected graph $G = (V, E)$ is a set $U \subseteq V$ s.t. every vertex $v \in V - U$ has at least one neighbour in U .

Thm 3: $G = (V, E)$ be a graph on n vertices, with min degree $\delta > 1$. Then G has a dominating set of at most $n \frac{1 + \ln(\delta+1)}{\delta+1}$ vertices.

Proof: $P(\text{an edge}) = p$. Let $X \subseteq V$ and no vertex in $V - X$ has neighbor in X . $P(v \text{ and its neighbors are not in } X) \leq (1 - p)^{\delta+1}$. Take X_v is an indicator r.v. where $X_v = 1$ if v has neighbor in X , $X_v = 0$ if v has no neighbor in X . Let $Y =$ vertices in $V - x$ which has neighbor in X . $E(|X| + |Y|) \leq np + n(1 - p)^{\delta+1} \leq np + ne^{-p(\delta+1)}$. To get maximum number of vertices we differentiate $E(|X| + |Y|)$ wrt p . We set $p = \frac{\ln(\delta+1)}{\delta+1}$. Then $E(|X| + |Y|) \leq n \frac{1 + \ln(\delta+1)}{\delta+1}$.

Lemma 1: $G = (V, E)$ has min degree δ , $V = V_1 \cup V_2$ be a cut smaller than δ . Then every dominating set U of G has vertices in V_1, V_2 .

Proof: Taking $U \subseteq V_1$ contradicts the size of the cut.

Proposition 2: Every n uniform hypergraph with less than 2^{n-1} edges is 2-colorable.

If the graph is colored red or blue and for simplicity take $\frac{v}{2}$ vertices colored red and $\frac{v}{2}$ vertices colored blue. $S \subset V$ is uniformly selected n -sets. Then $P(S \text{ is monochromatic}) \geq p = \frac{2(\frac{v}{2})}{\binom{v}{n}} =$

$$2^{1-n} \prod_{i=0}^{n-1} \frac{v-2i}{v-i} \sim 2^{1-n} e^{-\frac{n^2}{2v}} \text{ as } v \gg n^{\frac{3}{2}} \text{ and } \frac{v-2i}{v-i} = 1 - \frac{i}{v} + O(\frac{i^2}{v^2}) = e^{-\frac{i}{v} + O(\frac{i^2}{v^2})}.$$

If $m(n)$ = minimum possible number of edges in hypergraph s.t. graph is 2-colorable. Then we can show that $m(n) < (1 + o(1)) \frac{eln2}{4} n 2^{2^n}$.

Def 3: A subset A of abelian group G is sum-free if $(A + A) \cap A = \phi$, i.e., there is no $a_1, a_2, a_3 \in A$ s.t. $a_1 + a_2 = a_3$.

Thm 4: Every set of n nonzero integers contains a sum-free subset A of size $|A| > \frac{n}{3}$.

Def 4: $d(F) = |\{(F', F'') : F', F'' \in F, F' \cap F'' = \phi\}|$

Thm 5: Let F be a family of $m = 2^{(1+\frac{\delta}{2})n}$ subsets of $X =$ set of n elements, where $\delta > 0$. Then $d(F) < m^{2-\frac{\delta^2}{2}}$

Erdos-Ko-Rado Theorem: For $0 \leq s \leq n-1$ set $A_s = \{s, s+1, \dots, s+k-1\}$ where addition

is modulo n . Then F can contain at most k of the sets A_s .

Using this thm, we get a bound on $|F|$ i.e., $\frac{k}{n} \geq P(A \in F) = \frac{|F|}{\binom{n}{k}} \Rightarrow |F| \leq \frac{k}{n} \binom{n}{k} = \binom{n-1}{k-1}$

3.3.2 Applications of Linearity of Expectation

- No. of fixed points in a permutation σ of $\{1, 2, \dots, n\}$ = 1. Take indicator r.v. $X_i = 1$ if $\sigma(i) = i$ otherwise $X_i = 0$. Then $E(X) = E(\sum_i X_i) = \sum_i E(X_i) = \sum_{i=1}^n \frac{1}{n} = 1$.
- There is a tournament T with n players and at least $n!2^{-(n-1)}$ Hamiltonian paths. Take indicator r.v $X_\sigma = 1$ if σ generates a hamiltonian path, otherwise $X_\sigma = 0$.
- $G = (V, E)$ be a graph having n vertices and e edges. Then G contains a bipartite subgraph with at least $\frac{e}{2}$ edges. Take $T, V - T$ are 2 sets of vertices and X_{xy} is the indicator random variable which has value 1 if x, y are connected by an edge, 0 otherwise. So $E[X_{xy}] = \frac{1}{2} \Rightarrow E[X] = \sum_{\{x,y\} \in E} E[X_{xy}] = \frac{e}{2}$.
- If G has $2n$ vertices and e edges then G contains a bipartite subgraph with at least $\frac{en}{2n-1}$ and if G has $2n + 1$ vertices then G contains a bipartite subgraph with at least $\frac{e(n+1)}{2n+1}$. Similar proof as previous.
- There is a 2-coloring of K_n with at most $\binom{n}{a} 2^{1-\binom{a}{2}}$ monochromatic K_a . Take a random coloring. Let X be the number of monochromatic K_a . $E[X] = \frac{2\binom{n}{a}}{2^{\binom{a}{2}}} = \binom{n}{a} 2^{1-\binom{a}{2}}$.
- There is a 2-coloring of $K_{m,n}$ with at most $\binom{m}{a} \binom{n}{b} 2^{1-ab}$ monochromatic $K_{a,b}$. Similar proof as previous.
- Let $v_1, v_2, \dots, v_n \in \mathbf{R}^n$, $|v_i| = 1 \forall i \in [n]$ then there exists $\epsilon_1, \epsilon_2, \dots, \epsilon_n \in \{-1, 1\}$ s.t. $|\epsilon_1 v_1 + \epsilon_2 v_2 + \dots + \epsilon_n v_n| \leq \sqrt{n}$ and also $\epsilon_1, \epsilon_2, \dots, \epsilon_n \in \{-1, 1\}$ s.t. $|\epsilon_1 v_1 + \epsilon_2 v_2 + \dots + \epsilon_n v_n| \geq \sqrt{n}$. Let $X = |\epsilon_1 v_1 + \epsilon_2 v_2 + \dots + \epsilon_n v_n|^2 \Rightarrow X = \sum_{i=1}^n \sum_{j=1}^n \epsilon_i \epsilon_j v_i \cdot v_j \Rightarrow E[X] = \sum_{i=1}^n \sum_{j=1}^n v_i \cdot v_j E[\epsilon_i \epsilon_j]$ when $i \neq j$ then $E[\epsilon_i \epsilon_j] = E[\epsilon_i] E[\epsilon_j] = 0$. When $i = j$ then $\epsilon_i^2 = 1$ so $E[\epsilon_i^2] = 1$ thus $E[X] = \sum_{i=1}^n v_i \cdot v_i = n$. Taking sqrt we get the desired result.
- Let $v_1, v_2, \dots, v_n \in \mathbf{R}^n$, $|v_i| \leq 1 \forall i \in [n]$. Let $p_1, p_2, \dots, p_n \in [0, 1]$ be arbitrary and $w = p_1 v_1 + p_2 v_2 + \dots + p_n v_n$ then there exists $\epsilon_1, \epsilon_2, \dots, \epsilon_n \in \{0, 1\}$ setting $v = \epsilon_1 v_1 + \epsilon_2 v_2 + \dots + \epsilon_n v_n$ then $|w - v| \leq \frac{\sqrt{n}}{2}$. $Pr[\epsilon_i = 1] = p_i$ and $Pr[\epsilon_i = 0] = 1 - p_i$. Set $X = |w - v|^2 = |\sum_{i=1}^n (p_i - \epsilon_i) v_i|^2 = \sum_{i=1}^n \sum_{j=1}^n v_i \cdot v_j (p_i - \epsilon_i)(p_j - \epsilon_j) \Rightarrow E[X] = \sum_{i=1}^n \sum_{j=1}^n v_i \cdot v_j E[(p_i - \epsilon_i)(p_j - \epsilon_j)]$. For $i \neq j$ $E[(p_i - \epsilon_i)(p_j - \epsilon_j)] = E[(p_i - \epsilon_i)] E[(p_j - \epsilon_j)] = 0$. For $i = j$ $E[(p_i - \epsilon_i)^2] = p_i(p_i - 1)^2 + (1 - p_i)p_i^2 = p_i(1 - p_i) \leq \frac{1}{4}$. So $E[X] \leq \frac{1}{4} \sum_{i=1}^n |v_i|^2 = \frac{n}{4}$.
- Let $a_{ij} = \pm 1$ for $1 \leq i, j \leq n$. Then there exist $x_i, y_i = \pm 1$, $1 \leq i, j \leq n$ so that $\sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i y_j \geq (\sqrt{\frac{2}{\pi}} + o(1)) n^{\frac{3}{2}}$. Let $y_1, y_2, \dots, y_n = \pm 1$ be selected independently and uniformly and set $R_i = \sum_{j=1}^n a_{ij} y_j$ and $R = \sum_{i=1}^n |R_i|$. Fix i . Regardless of a_{ij} , $a_{ij} y_j$ is ± 1 with probability $\frac{1}{2}$ and their values are independent. R_i has distribution S_n - the distribution of the sum of n independent uniform $\{-1, 1\}$ random variables - and so $E[|R_i|] = E[|S_n|] = (\sqrt{\frac{2}{\pi}} + o(1))\sqrt{n}$. Using elementary calculus, alternative closed form $E[|S_n|] = n 2^{1-n} \binom{n-1}{\lfloor \frac{n-1}{2} \rfloor}$ may be derived combinatorially and the asymptotics follows from Stirling's formula. Now apply Linearity of Expectation to R : $E[R] = \sum_{i=1}^n E[|R_i|] = (\sqrt{\frac{2}{\pi}} + o(1)) n^{\frac{3}{2}}$. There exist $y_1, y_2, \dots, y_n = \pm 1$ with R at least this value. Finally, pick x_i with the same sign as R_i so that $\sum_{i=1}^n x_i \sum_{j=1}^n a_{ij} y_j = \sum_{i=1}^n x_i R_i = \sum_{i=1}^n |R_i| = R \geq (\sqrt{\frac{2}{\pi}} + o(1)) n^{\frac{3}{2}}$.

Chapter 4

Evaluation

Chapter 5

Conclusion

If we care about communicating in a minimum number of bits, these ideas are clearly fundamental. If we care about compressing data, information theory addresses the core questions and gives us the fundamentally right abstractions. But what if we don't care – are they anything other than curiosities?

Ideas from information theory turn up in lots of contexts: machine learning, quantum physics, genetics, thermodynamics, and even gambling. Practitioners in these fields typically don't care about information theory because they want to compress information. They care because it has a compelling connection to their field. Quantum entanglement can be described with entropy. Many results in statistical mechanics and thermodynamics can be derived by assuming maximum entropy about the things you don't know. A gambler's wins or losses are directly connected to KL divergence, in particular iterated setups.

Information theory turns up in all these places because it offers concrete, principled formalizations for many things we need to express. It gives us ways of measuring and expressing uncertainty, how different two sets of beliefs are, and how much an answer to one question tells us about others: how diffuse probability is, the distance between probability distributions, and how dependent two variables are.

Probabilistic method gives us the ways to solve combinatorial problems. Also information theory helps us to solve problems and coding theory tells us some special maps which transmits information correctly. These topics are co-related with each other can solve much real life problems.

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